

Identification and monitoring of livestock with biologging for EU policy implementation

A review

Cagnacci, F.

2026



This publication is an External Study report prepared for the Joint Research Centre (JRC), the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The contents of this publication do not necessarily reflect the position or opinion of the European Commission. Neither the European Commission nor any person acting on behalf of the Commission is responsible for the use that might be made of this publication. For information on the methodology and quality underlying the data used in this publication for which the source is neither Eurostat nor other Commission services, users should contact the referenced source. The designations employed and the presentation of material on the maps do not imply the expression of any opinion whatsoever on the part of the European Union concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

Contact information

Name: Francesca Cagnacci

Address: Fondazione Edmund Mach, via Mach 1 38089 San Michele all'Adige, Trento, Italy.

Email: francesca.cagnacci@fmach.it

Joint Research Centre

<https://joint-research-centre.ec.europa.eu>

JRC146300

PDF ISBN 978-92-68-41157-5 doi:10.2760/8082677

KJ-01-26-282-EN-N

Luxembourg: Publications Office of the European Union, 2026

© European Union, 2026

Some content was created using the support of AI tools for literature search, Consensus <https://consensus.app/>; Elicit <https://elicit.com/>; Google Scholar Lab https://scholar.google.com/scholar_labs/search?hl=en.



The reuse policy of the European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Unless otherwise noted, the reuse of this document is authorised under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence (<https://creativecommons.org/licenses/by/4.0/>). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of photos or other material that is not owned by the European Union permission must be sought directly from the copyright holders.

Cover picture: Cows on German sleeper dike, Uberprutser, CC BY-SA 4.0, via Wikimedia Commons

How to cite this report: Cagnacci, F., *Identification and monitoring of livestock with biologging for EU policy implementation - A review*, Publications Office of the European Union, Luxembourg, 2026, <https://data.europa.eu/doi/10.2760/8082677>, JRC146300.

Contents

Abstract	2
Acknowledgements	3
1. Introduction	4
2. Policy requirements for livestock animals' identification and tracking.....	6
2.1. Current systems for livestock animals' identification: from tags to digital platforms.....	6
2.2. Further challenges and opportunities of livestock farming: is there an added value from livestock movement tracking?.....	8
2.3. Restrictions for methods to track livestock animals' movement related to policy requirements	12
3. Technical solutions adopted for monitoring animals with biologging	14
3.1. What is biologging? Requirements and solutions.....	14
3.2. Biologging of wild terrestrial mammals for identification, detection, movement tracking and behavioural assessment: a review	18
3.3. Biologging: new perspectives and opportunities.....	20
3.4. Biologging applied to farm animals: Precision Livestock Farming vs Precision Livestock Extensive Farming	20
3.5. Scaled-up biologging applications to extensive or hybrid farming: data management and preservation caveats.....	30
4. Feasibility and scalability of biologging applications for tracking livestock movement in the EU policy context.	31
4.1. Potential technical solutions for tracking livestock movements and behaviour vs EU policy requirements	31
4.2. Challenges for upscaling technical solutions at EU level: data transmission, welfare, cost.	32
5. Opportunities and perspectives for the implementation of smarter farming (smart pastures)....	34
5.1. Possible approaches for turning technological opportunities into pragmatic solutions	34
5.2. Opportunities for coordination with existing initiatives	35
6. Conclusions.....	38
References	39
List of abbreviations and definitions	46
List of figures.....	48
List of tables.....	49

Abstract

This report evaluates how biologging (electronic devices deployed on individual animals that record movement, behaviour and physiology) can support livestock identification and traceability in the framework of existing EU regulations. The EU livestock sector accounts for 41 % of agricultural output (€218.8 billion in 2024). To maintain high food safety standards and respond rapidly to animal diseases, Regulation (EU) 2016/429 and Delegated Regulation 2019/2035 require permanent physical identifiers (ear tags, transponders) for certain terrestrial livestock species. Current systems already ingest data at the national level into animal registries part of the Integrated Administration and Control System (IACS) that manage the common agricultural policy (CAP) financial support. The data can feed into cross-national systems (TRACES) for intra-European Economic Area (EEA) trade. However, they lack real-time location data.

A range of biologging technologies developed for wildlife, for example Global Navigation Satellite System (GNSS) collars or ear tags, accelerometers, Internet of Things (IoT)/Low-Power Wide-Area Network (LPWAN) transmitters, can provide live-tracking, behavioural monitoring and physiological sensing (e.g. ruminal bolus temperature).

Mapping these sensors to legal categories shows that electronic ear tags equipped with GNSS or low-power IoT satisfy the 'additional' status for cattle, pigs and adult sheep/goats, with identification associated with localisation, while equines remain excluded by the possibility to use ear-tags according to current regulation.

The review of scientific literature and commercial products related to Precision Livestock Extensive Farming (PLEF) demonstrates maturity of these technologies and market readiness. This study highlights how biologging can contribute to four of the EU policy objectives reported in the Vision for Agriculture and Food (European Commission, 2025) and in the Strategic Dialogue on the future of EU agriculture (European Commission, 2024): 1) One Health (via precise mapping of grazing pressure and disease hotspots), 2) Animal welfare (by early detection of stress or injury via accelerometry), 3) Innovation (by 'self check-in' of animals, the administrative load is reduced), 4) Reward (by supporting effective checking and verification of grazing patterns, which could lead to policy and market remunerations).

Key challenges are data transmission (satellite offers global coverage but high cost; LPWAN is cheap but patchy) including powering, connectivity and interoperability, compliance with the EU General Data Protection Regulation (GDPR) and broader aspects of data ownership, data standardisation for integration with IACS, and acceptance of these new technologies (and the additional costs that these technologies may entail) by farmers.

Integrating GNSS/IoT ear tags seems to be the most policy-compatible, cost-effective pathway to modernise livestock monitoring, provided that EU-wide data-infrastructure for data transmission and standards are established, and that unit cost is minimised or supported.

Pilot activities, possibly leveraging the several ongoing initiatives on these topics, are needed to test the technical, operational and financial feasibility of an extensive application at EU level, including the creation of infrastructures for data transmission and management, and involving farmers from the beginning.

Acknowledgements

The author sincerely thanks Ferdinando Urbano (JRC), Nicola Britta Holsten (DG AGRI), Rudolf Span (DG AGRI), Valerio Abbadessa (DG AGRI), Francesca Chiodi (DG AGRI), Francesco Iadecola (DG AGRI) and Michele Nori (DG AGRI) for their thoughtful reviews and discussions, which greatly enhanced the quality of this work.

1. Introduction

The European livestock sector represented the 41.1% economic output of the entire agricultural sector in the European Union (EU) in 2024, corresponding to a total value of €218.8 billion and approximately €100 billion added gross value (after costs; source EUROSTAT, 2026). The sector is supported by the guidance of the EU Commission Directorate-General for Agriculture and Rural Development (DG AGRI), that aims at promoting a knowledge and evidence-based EU agriculture and food systems, stemming in innovation. One of the main actions to this end is the development, implementation, monitoring and evaluation of the Common Agricultural Policy (CAP). The CAP implementation is supported by the national Integrated Administration and Control System (IACS) used to manage CAP support to farmers. IACS includes a database of georeferenced data for the area-based agricultural sector and uses sectorial databases and identification for domestic livestock.

Currently, the livestock sector relies on a robust system of traceability to maintain high food safety standards and respond rapidly to animal diseases, with further guidance from the Directorate-General for Health & Food Safety (DG SANTE). The basic rules of this system are provided by the EU Council Regulation 2016/429 and subsequent delegated Regulations like Delegated Regulation (EU) 2019/2035, that establish the legal framework for physical identification of livestock animals (such as ear tags; see Chapter 2). Currently, there is an increasing shift toward digitalization and electronic identification to automate data capture and integrate on-farm systems with national official Identification and Registration (I&R) databases ([EU Livestock Workstream](#)). While for area-based CAP interventions use of satellite data, geo-tagged photos etc for monitoring already reduces control and administrative burden for farmers and national administrations, this is not used in animal related CAP interventions.

In this context, this report will evaluate the opportunity of the use of biologging, or the use of electronic devices attached to animals for live tracking of movement and collection of data on behaviour, and physiology, to support innovation in livestock identification. Biologging has been utilized in scientific research for over 80 years: this report evaluates how these technologies can be transitioned from biodiversity monitoring to the livestock sector to encourage systems that are cost-effective and fair. To this end, the primary purpose of this report is to detect technological gaps and define feasible implementation options for livestock animal monitoring solutions, while considering the current legal requirements. To achieve this, the working approach involved three distinct steps:

- assessment of policy requirements: a study of relevant regulations and subsequent synthesis of legal and innovation requirements, also in considerations of the interviews occurred with experts from DG AGRI (Chapter 2);
- bibliographic review: an evaluation of technical biologging solutions adopted in both wild animal research and agricultural contexts, specifically focusing on Precision Livestock Extensive Farming (PLEF) (Chapter 3);
- gap analysis/opportunities and perspectives: a matching of existing technical options with policy requirements to highlight issues related to economic viability, data management, and personal data protection, and analysis of opportunities and challenges for their implementation (Chapters 4 and 5).

By reviewing existing sensors—such as Global Navigation Satellite System (GNSS), usually Global Positioning System (GPS), tags, and accelerometers—and data transmission networks, this report

identifies possible pathways for modernizing livestock identification while respecting the specific management needs of different species, including the regulatory and legal framework, given substantial investment in a centralized infrastructure for data transmission.

Since the reviewed material is complex and rich, a choice was made to summarise it in tables and graphic elements, that represent an integral part of this document and are essential to correctly interpret the report.

2. Policy requirements for livestock animals' identification and tracking

2.1. Current systems for livestock animals' identification: from tags to digital platforms

The EU hosts a vast livestock population. By the end of 2024, approximately 132 million pigs, 72 million bovine animals, and 67 million sheep and goats were registered across the EU (EUROSTAT, 2025a). Given the European single market's free movement of goods, the presence of international commercial partners, and the intricate supply chains for products of animal origin, robust traceability is fundamental. To manage this complex network effectively, the EU enforces a comprehensive system of **identification** and **registration** based on the Animal Health Law (Regulation (EU) 2016/429). The primary objective of this framework is to ensure full traceability, a prerequisite for maintaining high food safety standards, enabling rapid epidemiological responses to transmissible animal diseases, and improving animal welfare. In addition, the CAP requires that the national IACS includes as a mandatory element the system for the identification and registration of certain species stemming from the sectorial veterinary legislation.

To achieve this, each animal must be individually identified through a permanent system from birth until slaughter or death. This physical identification is the cornerstone of the system, ensuring that every animal can be linked to its establishment of origin. Commission Delegated Regulation (EU) 2019/2035 of 28 June 2019, supplementing Regulation (EU) 2016/429 of the European Parliament and of the Council clearly delineates the specific means required for each category of relevant domestic animals. **Table 1** provides a detailed breakdown of these requirements, illustrating both distinct animal-specific approaches and a unifying trend toward digitalization. While conventional ear tags remain the baseline for most livestock (e.g. bovine, porcine, ovine and caprine animals), the regulation **tailors** the identification requirements to specific management needs and biological/exploitation characteristics of each livestock species. These range from the mandatory double-tagging of ruminant animals to streamlined protocols for species with shorter production cycles (such as porcine animals), and the use of injectable transponders as the primary identifier for equines to preserve their aesthetic value and to avoid possible injuries. Crucially, the regulatory framework increasingly accommodates **electronic identification** across all groups (**Table 1**). Whether mandatory or permitted as an alternative, this flexibility offers a significant opportunity to modernize livestock management. By leveraging these authorized electronic identifiers, operators can **automate data capture and seamlessly integrate on-farm systems with national animal registries**, thereby simplifying the procedures on the longer run, reducing human error and intervention, and enhancing the speed of traceability once specific platforms and protocols are set in place, which might require some initial investments.

Currently, movements of animals between Member States and certain EEA countries, as well as imports of live animals from third countries are monitored via **TRACES** ([TRAdE Control and Expert System](#)), which digitizes sanitary certification and ensures that only compliant animals enter the supply chain, thus representing an interoperable network of national databases and EU/EEA-wide digital platforms. Livestock species-specific networks also exist as an extension of the TRACES system, such as **BOVEX**, with the goal of simplifying and streamline the flow of information during bovine movements. This seamless integration of physical identifiers and digital monitoring currently available at EU/EEA level is critical not only for EU internal market stability but also for monitoring imports from non-EU countries, supporting the strategic autonomy and sustainable competitiveness

of EU agriculture. These initiatives built on the common goals of DG SANTE, fighting animal diseases with health control of live animals and products of animal origin ('prevention is better than cure'), and DG AGRI to support the transition to a more sustainable and competitive livestock system with positive effects on the management of European grassland.

Table 1. Authorized means of identification for kept animals under Commission Delegated Regulation (EU) 2019/2035, with specific derogations available.

Animals	Identification requirements	Derogations	External electronic identifier
Bovine <i>Art. 38</i>	Two conventional ear tags, one attached to each ear flap from birth*	One conventional tag can be replaced by an approved electronic identifier. Both tags can be replaced by an electronic identifier only under specific exemptions [#] (Art. 39).	Permitted
Ovine & Caprine Slaughter <12 months <i>Art. 45(1)</i>	At least one conventional ear tag OR conventional pastern band, from birth*	Exemption: One electronic ear tag is sufficient for transport to slaughter (via assembly or fattening operation in another establishment) within the same Member State (Art. 46(4)).	Permitted
Ovine & Caprine General >12 months <i>Art. 45(2)</i>	One conventional ear tag AND a secondary electronic identifier (points (c) to (f) of Annex III) from birth*	One conventional pastern band shall replace the conventional ear tag if the ears are too small (Art. 46(1)). If staying within the same Member State: - One tattoo may replace the conventional ear tag if the use of ruminal bolus is authorised (Art. 46(2)) - One tattoo may replace the electronic identifier (Art. 46(3)) - One ruminal bolus or an injectable transponder can be used instead only under specific exemptions [#] (Art. 47) - One conventional ear tag or pastern band may replace the electronic identifier if the Member State has a small registered population (<600 000 sheep/goats or <160 000 goats) and the animals are not intended to be moved to another Member State (Art. 48)	Compulsory, excl. exceptions
Porcine <i>Art. 52</i>	One conventional ear tag OR electronic ear tag OR tattoo from birth*	No identifiers if other means of traceability are in place and the animal does not exit the supply chain within the Member State (Art. 53). One injectable transponder can be used instead only under specific exemptions [#] (Art. 54).	Permitted

Animals	Identification requirements	Derogations	External electronic identifier
Equine <i>Art. 58</i>	One injectable transponder from birth	One conventional ear tag* can replace the injectable transponder if animals are kept for meat production and are born in or imported to that Member State (Art. 59). Specific populations living in semi-wild conditions may be exempted from identification until removed from the wild or brought into domestic use (Art. 60), Afterwards identification according to Art. 58. Alternative methods can be used only in exceptional cases (e.g. specific breeds/lineages, animal health/welfare) provide at least the same guarantees as the injectable transponder (Art. 62).	No specific provisions laid done. [§]
Camelidae <i>Art. 73</i>	Two conventional ear tags, one attached to each ear flap, OR Injectable transponder from birth*		Possible (if added or integrated to the conventional identifiers).
Cervidae <i>Art. 73</i>	Two conventional ear tags, one attached to each ear flap, OR Injectable transponder, OR tattoo from birth*	Member States may authorise a specific identification regime for reindeer kept in semi-wild conditions (Art. 74) considered as farmed game meat.	Possible (if added or integrated to the conventional identifiers).

Source: compiled by the author based on an assessment of the relevant regulation.

Notes: Means of Identification (Annex III) are (a) Conventional ear tag; (b) Conventional pastern band; (c) Electronic ear tag; (d) Ruminal bolus; (e) Injectable transponder; (f) Electronic pastern band; (g) Tattoo; (h) Leg-ring. * The identification code must be visible, legible and indelible, and can't be removed, modified or replaced without the permission of the competent authority. # For cultural, recreational, or scientific purposes and for mountain pastures. § Current regulation neither refers to the possibility of fitting an external electronic identifier, nor to the fact that it is forbidden.

2.2. Further challenges and opportunities of livestock farming: is there an added value from livestock movement tracking?

One of the most recent initiatives to promote the livestock sector 'towards the future' is the [EU Livestock Workstream](#), as part of the [Vision for Agriculture and Food](#) (European Commission, 2025). While the livestock sector is key to EU socio-economy (in terms of value and jobs), there are also increasing challenges, such as the spread of diseases due to globalisation, which is already well addressed by the current system (see 2.1); but also the uncertainties of production imposed by climate change; or the decrease of farms and farming in terms of attractiveness of working conditions, despite the increasing world demand of animal products. Some of these aspects have been analysed in the Vision for Agriculture and Food (European Commission, 2025) and in the Strategic Dialogue on the future of EU agriculture (European Commission, 2024), suggesting some pillars to promote a new generation of farmers and farming activities related to livestock:

- adopt a **One Health** approach, having sustainability and quality of production at the core, including organic production, agroecology, agroforestry, regenerative agriculture, pastoralism;
- consider **animal welfare** as part of the game, as increasing welfare means decreasing disease susceptibility and greenhouse gases emissions (CH₄, NO₂);
- ease up the adoption of innovation, and particularly **digitalisation**;
- make the recognition and **reward** for sustainable practices substantial and available, so promoting their adoption by farmers, and promoting farming jobs among citizens, while **decreasing the administrative load**.

Extensive/hybrid farming and **pastoralism** are practices that are well placed in this context. Pastoralism for example is recognised as a deeply culturally and historically valuable practice in Europe: 2026 is the International Year of Pastoralism, and European stakeholders see this as an opportunity to push for a European Action Plan for pastoralism ([Euromontana.org](https://euromontana.org)). Key priorities would be the improvement of the working conditions of shepherds and traditional farming, animal welfare and health, extensive farming resilience to climate change, limitation and mitigation of impacts of husbandry (with preservation of ecosystem services and biodiversity, see One Health concept above).

Most importantly, the valorisation of extensive farming/pastoralism-derived products (a 'labelling' or 'branding') are essential requirements to reward the challenges of venturing in these alternative practices (*'to be preserved while preserving'*). Exposure to recolonising large carnivores is also a major concern of stakeholders.

From a sustainability perspective, farming is certainly a human activity which is responsible for a considerable amount of greenhouse gases. It is estimated that it accounts for 12% of EU emissions (EUROSTAT, 2025b). The aforementioned EU Livestock Workstream helps understand the trade-off between intensive farming, with high land-use efficiency, but disproportionate need of feed and amount of emissions, and extensive farming that contributes to maintaining ecosystem services and carbon storage, but requires more land and where mitigation strategies are more difficult to adopt. The comparison between intensive and extensive systems is complex and dependent on various factors, including management practices, regional conditions, and technological applications in both types of systems. Critically, the number of animals per unit of land is a key value that could unbalance the trade-off, for example making manure management either a resource for natural (i.e. not industrial produced) nitrogen supply, or an overload of the same chemicals in the ecosystem with cumulation in drainage water (nutrient runoff; Oquist et al., 2007; Holbrook, 2012; Leolini et al., 2023). The CAP can adapt to this multi-faceted and complex socio-economic and environmental context with technical measures that **encourage sustainable livestock systems and are cost-effective and fair**. Fair, in this context, means to support farmers, but also to provide real choices to consumers (for example, with a resilient food system, and transparencies and authenticity of labels).

Current EU CAP Network's Thematic Groups¹ are suggesting technical innovation, focussing on the CAP's role in i) checking the implementation of EU farming policy and regulations while ii) promoting

¹ See for example https://eu-cap-network.ec.europa.eu/thematic-groups-cap-implementation/thematic-group-caps-role-supporting-sustainable-and-competitive-livestock_en

a fair and inclusive transition to more sustainable, competitive and diversified agri-food systems ([EU CAP NETWORK](#)). The adoption of innovative tools in the identification and monitoring of animals would help move the livestock system **from compliance to performance**, as it is being already implemented with the aid of Earth observation, particularly the Copernicus satellites Sentinel 1 and 2 ([copernicus.eu](#)), for area-based interventions in the CAP 2023-2027 through the Area Monitoring System (AMS), a new IACS component introduced in the current CAP ([Tools4CAP](#)).

However, a sustainable, alternative, workers-centered farming system could rely on a centralized identification and tracking of livestock to promote the quality and uniqueness of these socio-economic models (**Table 2**). An important contribution in this direction would be the introduction of innovation-based approaches to the well-established need for identification and tracking of livestock animals, where animals are **not only permanently identified, but also live-monitored and tracked (Table 2)**. The potential advantages of live-tracking animals reside in the possibility to link farming activities to the spatial distribution of animals and their interaction with the environment.

In other words, live tracking allows not only to follow the occurrence of livestock and its movements, or along the production chain, as currently required; but also to explicitly account for **the use livestock makes of land**. This is innovative and critical, because it allows farmers and policy to evaluate the trade-offs between number of animals per unit of land, production and sustainability not in abstract terms, but directly derived from animals' use of land, preserving the ecosystem value of livestock grazing (source of nutrients, renovation of grassland), while not overloading habitats (Strategic Dialogue on the future of EU agriculture **Objective 1**, see **Table 2**).

Similarly, when respected, a balanced land use by farm animals assure their health and welfare (**Objective 2**), providing clear evidence for the 'good for the environment, good for animal welfare' sweet spot (Broom 2019). Importantly, live tracking can be associated with other sensors that reveal energy expenditure vs food intake (Bailey et al., 2018 for a review; see Chapter 3.4). Further, the adoption of live-tracking animals would provide innovation and make the identification and tracking of animals even smoother (**Objective 3**), decreasing the administrative load through a strong increase in digitalisation (individual animals would be 'self-checked-in').

This requires an initial effort to establish new practices and inform farmers on the use of these technologies. This will need further investigation to ensure the acceptance by farmers of these innovative approaches, including the additional costs that these technologies may incur, which might require financial support to ensure that they do not perceive this as an additional administrative burden (Chapter 5).

Finally and crucially for a fair sustainable livestock system, live-tracking strongly support the verification of practices supported by the CAP (**Objective 4**), that would incontrovertibly be shown, so indicating (i) the transparency of application of alternative sustainable practice, such as hybrid, extensive or pastoralism-based farming; (ii) providing farmers with highly valuable branding; (iii) and consumers with choices that correspond to real practices. Another possible advantage of farming live-tracked animals would be to increase the interest for farming jobs by young generations that would be attracted by innovation, environmental sustainability, and societal trust.

CAP implementation would benefit from these innovative systems (**Table 2**) since farming practices would be effectively monitored with a high degree of automation, with no need for additional controls, covering all EU farms applying for concerned CAP support. As a result, the Vision for Agriculture and Food (European Commission, 2025) would be supported while creating a stronger link between the needs of producers, stakeholders, and EU policy (**Table 2**). For example,

one opportunity would be the integration with the results of AMS for monitoring low-intensity grazing, since this practice is difficult to detect with Satellite data.

Table 2. Potential added value of livestock movement tracking with biologging devices mapped against the objectives defined in the Strategic Dialogue on the future of EU agriculture.

Objective	Impact on CAP	Stakeholders/ extensive farming	Stakeholders/ pastoralism
One Health	Increased ability to support and monitor sustainable practices (hybrid and extensive farming, pastoralism, grassland management).	Clear understanding of the use of land by animals and economic sustainability of the practice (e.g. grazing vs renovation).	Better tracking of the herd both in transhumance and in the resident season. Better understanding of the use of pastures. Monitor interaction between livestock and wildlife to assess and mitigate risks of disease spill-over and predation. Gaining insights into traditional migration paths, enabling optimization for improved resource utilization and reduced environmental impact.
Animal welfare	Same advantages of current monitoring; faster identification of critical cases and their spatial origin.	Identification of suspect behaviors, or isolation from the herd. Monitoring of grazing vs activity.	Identification of suspect behaviors, or isolation from the herd, preserving welfare in challenging environments and movements. Monitoring of grazing vs activity. More precise traceability of contacts among animals and among establishments in case of control of animal diseases with possible reduction of the number of affected establishments and killed animals.
Innovation	Increased digitalisation and smoother linkage between farmers and policy. Use of data by farmers for sustainable farm management	Decrease administrative burden by animals 'self-checking in'. Possibility to monitor the herd remotely. Improvement of working conditions.	Decrease administrative burden by animals 'self-checking in'. Possibility to monitor the herd remotely. Improvement of working conditions. Preserving cultural heritage while enhancing efficiency without compromising traditional methods. Optimisation of farming processes linking agricultural machinery to enhance the quality and efficiency of feed production.
Reward	Monitoring allows for a preventive approach reducing penalties to farmers compared to ex post controls. Trustfulness of 'sustainability branding'.	Sustainability branding which is trusted and uniquely associated with quality. Reward of choices.	Sustainability branding which is trusted and uniquely associated with quality. Reward of choices. Preservation of traditional pastoral practices and traditional routes, contributing to the socio-cultural aspects.

Source: compiled by the author (see notes for references to the websites used to collect information).

Low intensity grazing systems are common in certain EU regions, for example in the Alpine and Mediterranean bioregions and in Western Ireland (De Arevalo and Van Der Velde, 2025) and are key components of many national CAP Strategic Plans. A live-tracking monitoring of livestock position and activity on grassland would be of great aid to develop and validate models for detection of grazing based on remote sensing data.

While the advantages of live-tracking animals to investigate their use of habitat and resources has been known in wildlife ecology for decades (Boyce and McDonald, 1999; Manly et al., 2002), with hundreds species and hundreds of thousands individuals being tracked and their tracks annotated with environmental variables (Kays et al., 2015; see Chapter 2.2), this aspect is still rarely considered for farming animals (e.g. Swain et al., 2011; Bailey et al., 2018; reviewed in Chapter 3.4), and mainly linked to either deployments of 'Virtual Fences' to monitor the permanence of individuals within unfenced farmland (Waterhouse 2023; e.g. [Nofence.com](https://www.nofence.com)), or fine-scale animal production studies for research purposes (e.g. Putfarken et al., 2008).

However, the opportunities offered by these sensors must be reconciled with further requirements that are key to their scalability: compliance with Policy requirements (see Chapter 2.3), data preservation (Chapter 3.4), effective simplification, cost-effectiveness and uptake by the farming community (Chapter 4).

2.3. Restrictions for methods to track livestock animals' movement related to policy requirements

Under the current regulations, the methods available to track livestock movements are strictly delineated by policy requirements rather than technological availability. Based on the aforementioned Commission Delegated Regulation (EU) 2019/2035 (**Table 1**), the use of **external identification devices** that would be reconcilable with live-tracking is only permitted in some specific cases and must meet rigorous technical specifications (i.e. being tamper-proof, unchangeable codes, and interoperability). Further, the approved electronic identifiers are primarily designed for identification and sanitary control rather than real-time location monitoring, therefore their use is possible with specific reading structure points (e.g. RFID) rather than continuous outdoor tracking.

Also, operators are currently not legally bound to deploy additional, non-standard, tracking technology (such as GNSS collars). Instead, they must adhere strictly to the approved means (e.g. external electronic identifiers such as electronic ear tags and pastern bands) that hold legal validity for official traceability and trade between Member States. To external identifiers, other means of allowed identification might be added that allow the use of additional sensors, for example ruminal bolus for bovines allow metabolic measures (see Chapter 3.4).

Within these regulatory boundaries, however, flexibility exists for operators to reconcile adherence to policy requirements with private herd management goals. **Table 3** illustrates these possibilities, distinguishing between contexts where electronic identification is the ordinary, compulsory standard (ORD) and where it can be voluntarily deployed in addition to permitted tags (ADD). The table also highlights sectors where external electronic identifiers are largely not permitted (NOT), such as in the equine sector. In most cases, electronic identifiers are permitted, but not compulsory, indicating that the digitalization of livestock tracking currently depends heavily on the farmers' intention to integrate innovative tags into their compulsory management system. Hence, on the long run it might be advisable to consider combining tracking functions with official identification to reduce the cost, as such combination ensures a high uptake.

Table 3. Regulatory status of external electronic identifiers (ear tags and pastern bands) by domestic animal species.

External electronic identifier	Bovine	Ovine <12 months	Ovine >12 months	Porcine	Equine	Camelid	Cervid
Electronic ear tag	ADD	ADD	ORD	ADD	NOT	NOT	NOT
Electronic pastern band	ADD	ADD	ORD	ADD	NOT	NOT	NOT

Source: Compiled by the author based on an assessment of the relevant regulation.

Notes: ORD: Ordinary, compulsory tagging, ADD: Additional, voluntary tagging, NOT: No special provisions in Commission Delegated Regulation (EU) 2019/2035

3. Technical solutions adopted for monitoring animals with biologging

3.1. What is biologging? Requirements and solutions

Biologging is a technical approach that consists in the use of electronic devices attached to an animal that collect biologging data (Davidson et al., 2025) on the movement, behaviour, physiology, or the external environment of the animal (Ossi et al., 2018; **Table 4**). These electronic devices typically include one or multiple sensors (Rutz and Hays, 2009) that automatically sense and measure animal-borne or environmental variables. Early deployments mainly occurred in the marine environment, where the inaccessibility to human eyes pushed the development of technology for animal monitoring able to function without human intervention. Typically, these early biologgers would last for short periods of time (hours or days) and ‘pop up’ on the ocean surface for retrieval (Rutz and Hays, 2009). When the opportunity to obtain information on animals’ lives directly from animals’ behaviour, i.e. not mediated by human observations, emerged in **terrestrial systems** (Cagnacci et al., 2010), other **challenges** were faced.

Deployment of biologgers does not only imply the use of a sensor, but also an attention to the specific goals and to the entire ‘life’ of the data, from collection to preservation. Hence, before deployment of biologgers it is fundamental to run a requirement analysis considering the goals and the context (**Table 4; Figure 1**).

First, multiple sensors progressively became available to typically track and classify animal movement on land, so a choice of the types of biologgers should consider the **motivation** for the deployment, as each sensor comes with a cost in terms of **battery consumption** and **data collection**, alongside the actual **tag cost**.

Second, **data shall be retrieved** for further processing and analysis, typically ‘remotely’, i.e. without direct contact with the device, especially for animals that are difficult to contact. Hence, in most cases data retrieval should rely on remote data transmission.

Third, **adjustments to data collection protocols** and schedule might be needed during the deployment, for example to monitor an animal at a higher frequency during a certain behavioural phase, or for management reasons (e.g. to monitor a problematic individual; to rescue a herd). This need corresponds to the establishment of a **two-way communication** between an operational centre (a server) and the deployed device.

Fourth, while data are typically retrieved in short, coded text messages, they should flow in a database, where they can be **processed, stored, and preserved**. The decisions on the alternatives for these requirements should consider the trade-offs in terms of complexity of the electronic components, the battery consumption, the durability of the casing, and the resulting single unit cost (**Table 4**). Importantly, the electronic components are ‘housed’ into some type of casing that is attached to the animal, and that should first and foremost respect its **welfare**, then assure **durability** and resilience to weather and other external stressors (e.g. animals kicking each other).

Each biologging deployment on terrestrial species aimed at identifying individuals and monitoring their movement and behaviour should consider the requirement analysis in **Table 4**. The main and most recent alternate scenarios offered by the combination of the decision elements are summarised in **Figure 1**. GNSS collars or ear tags equipped with a **GNSS sensor** use GNSS

satellites to triangulate the positions that in a sequence provide the movement trajectory or ‘track’ of an animal within a certain environmental context. The position determined by the sensors must then be transmitted to the farmer or to a centralised repository to be used. This information can be from sensors retrieved either as text messages sent through the **GSM network** (not represented in the figure), or via dedicated **commercial satellites** (e.g. Iridium). Notably, recent systems such as the new generation of ICARUS (see below Chapter 3.3) use dedicated ad hoc **small satellites**. Alternatively, data can be retrieved via radio sensors organised in an **Internet of Things** network (IoT), where regional systems of antennas can ‘bounce’ from one to the other and finally to the Internet the information transmitted by the tags. These systems are *de facto* **Wireless Sensor Networks (with low power communication, LPWAN)** that works two-ways, so usable to reprogramme the tags, and potentially also providing the tag location by measuring parameters of reception (such as Received Signal Strength Indicator, RSSI). Local versions of this approach use **UHF (Ultra High Frequency) radio transmission** with this purpose, where mobile units are associated with fixed units to locate the mobile ones and download the data (base stations, in this case). Typically, the UHF units can also identify one another within a certain RSSI range, operating as **sensors of proximity** between animals equipped with the tags. Alternatively, specific radio frequencies emitted by loggers can be used by central antennas to triangulate them, posed that all radio units are precisely temporised (typically through a GNSS sensor, used in this case to synchronize the time and not directly to triangulate locations, **reverse-GNSS**).

Table 4. Requirement analysis for the deployment of biologgers. Literature references in Notes.

Requirements	Types	Use	Pros	Con
Loggers/Sensors	GNSS loggers	Detection, Location	Automatically triangulates location. Can be used to set up virtual fences ² . Can be associated with other sensors.	Needs a transmission system to retrieve data live. Cost. Battery life.
	UHF loggers	Detection, Location	Inexpensive. Can be associated with other sensors.	Location needs to be extrapolated by an on-site Wireless Sensor Network (either using Receiver Strength Signal Indicator ³ or signal arrival time ⁴).
	Accelerometers	Activity, energy budget-extrapolated health status/fatigue ¹	Can be associated with other sensors. Provides innovative indices of health.	High memory and battery consumption. Impossible to remotely transmit given the load of data.
	Physiological sensors	Physiological/health status	Key parameters for health status, can be used within ruminal bolus.	Normally cannot be remotely transmitted. They depend on the shape and positioning for accurate data collection.

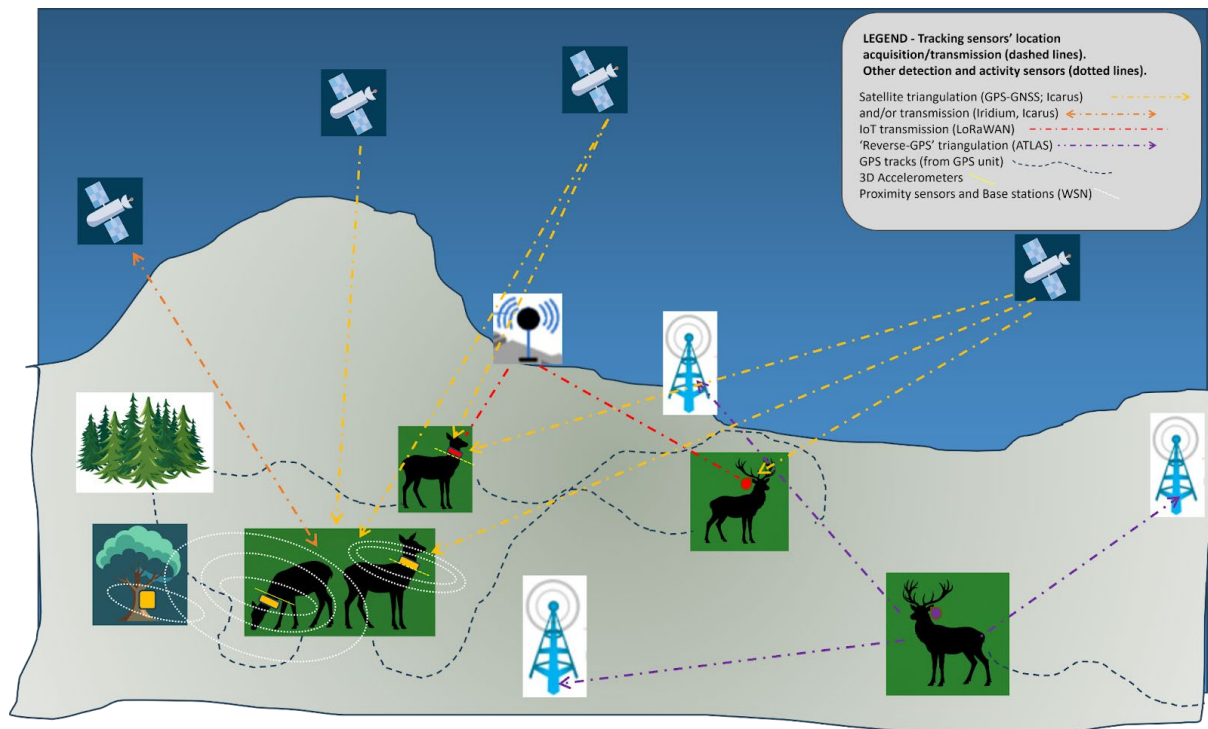
Requirements	Types	Use	Pros	Con
Data transmission	GSM network	Works with a SIM card inserted in the logger.	Inexpensive and reliable with GSM coverage. It allows fine tuning of the transmission frequency. It allows two-way communication.	Limited to areas covered by GSM. The shift between different systems (4G, 5G) has decreased the interest of companies to use this transmission method, in search of a more stable system.
	Pay-per-locations satellites (IRIDIUM)	Data are sent to a commercial satellite that charges per frequency of the locations.	Global coverage. It allows fine tuning of the transmission frequency and packages.	Very expensive, only relatively reliable.
	Dedicated mini satellites (e.g. CuboSats)	New systems increasingly utilize dedicated constellations of small satellites, such as ICARUS with CuboStats.	It shall support data retrieval and seamless transmission to supported database and/or mobile apps.	Not many applications yet.
	IoT (UHF loggers ⁵ ; Low-Power Wide-Area Network (LPWAN) ⁶)	IoT networks such as base stations of UHF loggers can store and transmit or dedicated LPWAN low power receivers can receive and transmit dataset.	No upfront costs once deployed. It can also lead to additional location estimates. It allows two-way communication.	It either works locally (UHF) or with regional systems (LPWAN-SigFox, LoRaWAN). In the first case, if animals exit the range of transmission they may be 'lost'.
Two-way communication	GSM network	Works with a SIM card inserted in the logger.	GSM communication allows to reset the loggers' schedule.	Limited to areas covered by GSM.
	UHF	Loggers can be reset either with base stations or hand-held units.	Several loggers with UHF units can receive logger reset.	UHF relies on a small transmission range, which may imply labour to find the animals to transfer data, decreasing the reliability.
	IoT ⁷	The LPWAN networks of antenna can reset the loggers in the range.	Low power IoT can reset loggers, also when data is otherwise uploaded by satellites saving energy (ICARUS).	IoT has a larger range, but it would 'lose' contact if animals exit the areas covered.

Requirements	Types	Use	Pros	Con
Data management platform	Seamless live-data transfer (e.g. Movebank ⁸)	Certain systems transfer data received from e.g. satellites (ICARUS) directly to data platforms that assure live-upload.	No additional labour in transferring the data, which are immediately preserved.	Data receive limited curtain before entering the database.
	Secondary upload (post-curation; e.g. Euromammals ⁹)	Data are typically otherwise downloaded on peoples' computers or companies' platforms, to then be stabilised in databases.	Data are highly curated before entering a database.	High labour is required to assure data quality standards.
Battery life	Battery life is a key concern and highly depends on several factors, such as: (i) the frequency of data collection (either programmed, for e.g. GNSS and accelerometers, or occurring, for e.g. proximity sensors); (ii) the coverage of signal (especially for GNSS); (iii) the frequency of data transmission. The requirement analysis for the needed battery should therefore consider the points above; as well as the desired duration, which in turns depends on the possibility to contact the animals to change or recharge the batteries; and, finally, the trade-off with the sensor unit rate- the longer the battery duration, the higher the weight. Solar-powered biologgers are a viable and cost-effective alternatives for animals mainly ranging in open areas with a relatively continuous light exposure (which depends on the latitude and season).			
Shape, size and weight	The shape and size and especially the weight should first and foremost be compliant with animal welfare, should that be for wildlife species or livestock. Weight depends on the miniaturization of technological components but mostly, as mentioned above, on the choice of battery. For large mammals, collars have been the favourite choice to relax weight restrictions. However, there has been a recent interest in ear tags, because of applicability to smaller species, animal welfare issues and the availability of cheaper options ¹⁰ , aligning better with livestock welfare requirements.			
Cost	The cost of sensor units depends in principle on technical components, yet most microelectronic boards and sensors are nowadays 'off-the-shelf', hence the energy optimisation and the efficiency of the packaging, as well as the opportunities offered by data transmission and data processing, highly influence the final cost. Another element is certainly the customisation and scalability of the units. Costs may therefore hugely vary. The typical price for a GNSS collar varies between €1 000-3 000, whereas IoT or UHF units are much cheaper, and as low as €100 under certain conditions.			

Source: compiled by the author.

Notes: ¹Delgado-Rozier et al., 2025; ²Lahoz-Monfort and Magrath, 2021; ³Rutz et al., 2015; ⁴Bijleveld et al., 2022; ⁵Picco et al., 2015; ⁶Wild et al., 2023; ⁷Krondorf et al., 2022; ⁸Kays et al., 2022 ⁹Urbano and Cagnacci, 2021; ¹⁰Stewart et al., 2025. Most acronyms referred to biologging sensors or related systems are summarised in the Glossary.

Figure 1. Biologging systems for animal detection and live tracking.



Source: redrawn and modified by the author from Wilmers et al., 2015.

3.2. Biologging of wild terrestrial mammals for identification, detection, movement tracking and behavioural assessment: a review

Table 5 provides a reasoned summary of the main biologgers available for terrestrial mammals, when the goal of the deployment is **identification, detection, movement tracking and behavioural assessment** of the tagged individuals. For completeness of information, further sensors on the physiology, health or reproductive status are also mentioned, as these can be combined with the identification/detection/positioning sensors. **Table 5 should be read against the alternate choices of the requirement analysis in Table 4**, with the main reason for deployment, together with the type of information obtained, that should guide the choice for the most appropriate sensor. Further, particular attention is given to the possibility to use the sensor for individual identification: any sensor not providing this possibility should be associated with others that do so.

Table 5. Main biologging sensors and technology for basic and applied studies of wild terrestrial mammals.

Technology and literature references	Biologger (B)/Remote sensor (Rs)	Information	Individual identification	Main application
VHF Radio Tags ¹	B/Rs	Position	Yes	Space use
GNSS Tags ²	B/Rs	Position	Yes	Space use
UHF-WSN Proximity loggers ³ ; IoT-LPWAN systems ⁴	B/Rs	Position, Detection, Contacts	Yes	Space use, Interactions
UHF-based passive triangulation systems (reverse-GNSS) ⁵	B/Rs	Position, Detection	Yes	Space use
PIT Tags (RFID) ⁶	B/Rs	Detection	Yes	Resource and/or space use
Photo identification ⁷	Rs	Detection, behavioral coding	Yes (if animal can be identified, e.g. by eartags)	Resource and/or space use/behavior
Aerial Platforms for Survey ⁸	Rs	Detection, behavioral coding	No, unless photo identification is in place (see above)	Species presence (census)/space use/behavior
Subjective cameras ^{9, 10}	B	Behavior	Yes	Behavior/Interactions
Accelerometers ¹¹	B	Activity	Yes	Activity patterns
Loggers for studying locomotion and navigation ^{12, 13}	B	Internal state	Yes	Physiology/movement
Jaw movements detectors ¹⁴	B	Jaw movements	Yes	Activity patterns/feeding behavior
Physiological state loggers ^{15, 16}	B	Internal state	Yes	Physiology/movement
Mortality sensor	B	Mortality detection	Yes	Demography/fitness
Vaginal sensor ¹⁷	B	Birth event detection	Yes	Demography/fitness

Source: compiled by the author, modified from Ossi et al. 2019.

Notes: ¹ MacDonald and Amlaner, 1980; ² Cagnacci et al., 2010; ³ Picco et al., 2015; ⁴ Wild et al., 2023; ⁵ Beardsworth et al., 2022; ⁶ Cagnacci and Massei, 2008; ⁷ Ryan et al., 2022; ⁸ Corcoran et al., 2021; ⁹ Kazcensky et al., 2019; ¹⁰ Ambris-Vilchis et al., 2015; ¹¹ Chimienti et al., 2016; ¹² Mendes et al., 2015; ¹³ English et al., 2024; ¹⁴ Mulvenna et al., 2018; ¹⁵ Wilmers et al., 2015; ¹⁶ ALzhal et al., 2015; ¹⁷ Barbnekt et al., 2009. These references are not exhaustive, but representative of the field.

3.3. Biologging: new perspectives and opportunities

Some elements are emerging in the biologging world that indicate an extreme dynamic and innovation potential in the field, at an even faster pace than what has been observed so far: **miniaturisation, distributed satellites, Internet of Things (IoT), drones and open sourcing.**

- As the field explodes, smaller and cheaper tags become available, contributing to the uptake of these technologies.
- Data retrieval is the limiting factor nowadays, and the one for which a faster innovation pace can be predicted. A combination of on-land and remote methods are particularly promising, such as the IoT networks, for their ability to combine triangulation, data transmission, and two-way communication (**Table 4**); drones, for opportunistic data retrieval; and systems of dedicated small satellites that can customarily transmit data, especially when combined with IoT-based communication systems to set up and control the tags. The version 2.0 of [ICARUS](#) is based on this principle.
- Open sourcing regards both hardware and software and is moving the market away from the classic scenario that was observed in the past, where many manufacturers tended to collapse into maximum two or three big players, with many companies now contributing to a faster innovation process and better integration of different technologies. While this is also due to the extreme specialisation of the sector, open sourcing has a big role in this, as it allows a closer connection between R&I and production.

3.4. Biologging applied to farm animals: Precision Livestock Farming vs Precision Livestock Extensive Farming

When Biologging is applied to farmed animals, it becomes part of the so-called **Precision Livestock Farming (PLF)** that has emerged and is typically used for housed animals (Berckmans et al., 2017). When deployment of biologging units pertains to livestock managed in rangeland, extensive or hybrid farming systems, it is referred to as **Precision Livestock Extensive Farming (PLEF)**; Bernabucci et al., 2025).

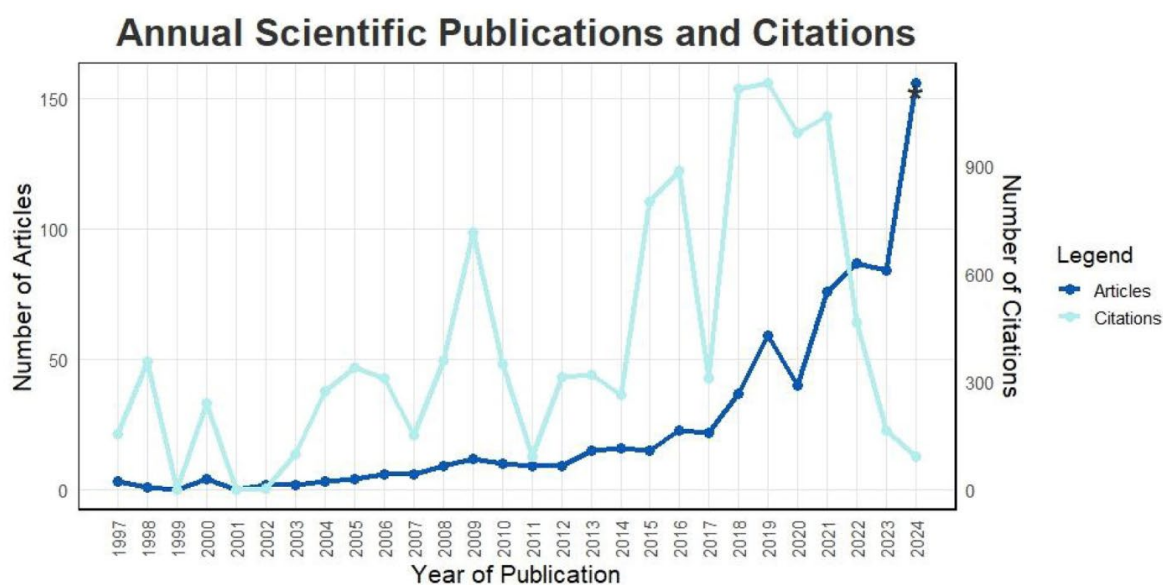
To explore the existing solutions and applications, a review of what is available in the scientific literature has been conducted together with an assessment of the solution available on the websites of commercial companies. To cover the diversity of applications existing or emerging, to later evaluate their suitability for tracking livestock in the EU policy context, a mixed systematic and expert-based approach has been applied:

- given the high number of classic systematic literature reviews recently published on the subject (see below; **Table 6**), a systematic keyword-based search in scientific literature databases has not been repeated, since the reported reviews have been leveraged to this end;
- systematic reviews have been complemented by AI based literature search, using natural language question-based search (question: 'Primary field studies deploying wearable tracking devices on free-ranging livestock') in three algorithms developed for scientific literature search (Consensus <https://consensus.app/>; Elicit <https://elicit.com/>; Google Scholar Lab https://scholar.google.com/scholar_labs/search?hl=en);
- further, a manual search based on papers' references has also been run to complement less obvious titles/keywords not included in the previous two methods;

— finally, websites of commercial products have been scrutinised.

After this search, a critical selection has been performed to cover the diversity of opportunities and criticalities for the deployment of biologging to track livestock, rather than a comprehensive list of all papers covering the subject (**Table 7**), also in consideration of the expert knowledge based on wildlife research (**Table 5**).

Figure 2. Recent uptake of literature on PLEF as reported by a recent review performed with systematic criteria within the period 1997–September 2024.



Source: Bernabucci et al., 2025.

Literature coverage of PLEF has grown extremely fast in the last 5 years (Bernabucci et al., 2025; **Figure 2**). Notably, the expansion of technological opportunities (hardware, software, analytical procedures) alongside the interest of scalable applications, has spurred a remarkable number of dedicated literature reviews that by themselves represent a vast source of information (**Table 6**).

Reviews considered the **classification of tag types in PLEF**, the **reasons why they are used**, but also, and possibly surprisingly, **the analytical methods to ‘make the best out of the tools’**. While this aspect was prevalent in wildlife ecology for what concerns free ranging animals, the opportunity to extract advanced information on farming productivity and farmed animals’ health has tremendously pushed the field, shortening the distance between PLEF and both PLF-housed animals and free-ranging wild species.

Despite literature ramping up in the PLF and PLEF domain (see **Table 6**), some common fields of interest can be identified (**Table 7**) when the available technology is analysed against the requirement analysis categories for deployment of biologgers (**Table 4**). Two core technologies are dominating the sector: **GNSS** and **tri-axial accelerometers**, which together comprise approximately 85% of deployed solutions in extensive livestock systems. While GNSS collars remain the standard for analysing **spatial distribution and virtual fencing efficacy**, accelerometers have become indispensable for **granular behaviour classification** (e.g. differentiating grazing from rumination). Most importantly, most papers do not only refer to the sensors used, but also to the **modality to transmit the data** (**Figure 3**, **Figure 4**). While this latter aspect is secondary in housed animals, it is **a key component for rangeland animals**, especially those farmed with a

pastoralism approach as the reduced contact is a key aspect to manage. Beyond continuous contactability, an important requirement of data transmission systems is a **seamless communication with centralised servers** (e.g. at the national level), where information on farmed animals is linked to the pertinent legal aspects (identification, origin, health statuses). Further, the field is currently transitioning from single-sensor deployments toward **multi-sensor fusion**—integrating heart rate, audio, and proximity sensors—to provide a holistic, real-time view of animal state in complex environments. From a species’ perspective, while cattle and sheep/goats remain the primary focus species, recent trends indicate a rising interest in free-ranging goats and outdoor swine.

Table 6. Recent reviews on biologging use in the PLF domain.

References	Review methodology	General output
Bailey et al., 2021	Non-systematic (perspective/commentary paper)	Potential outputs obtained from tracking rangeland animals with GNSS and accelerometers, including: monitoring of animals’ positions; distance travelled; management of grazing distribution and quantity; behavioural patterns; activity patterns; health status and welfare.
Lee & Seo 2021	Systematic, keyword-based Scopus database review. AND Review of WWSSs available in the market	Classification of PLF sensors to monitor cattle (named - wearable wireless sensor systems, WWSS): - motivation: ‘on-animal’, ‘near-animal’, and ‘from-animal’ (integrated into data loggers directly attached to the animal; positioned in the immediate vicinity, within farm structures; collect data indirectly); - types of sensors: GNSS, RFID, pressure sensors, intra-ruminal sensors, intra-vaginal devices, environmental sensors, microphones, MAG, GYR, cameras and thermal cameras, thermometers, heart rate).
Aquilani et al., 2022	Multi-step literature review based on Google Scholar and Scopus databases, without any limits on the data, country, or climate.. AND Review of sensors available in the market	PLF applications in pasture-based systems (PLEF): - animal identification (RFID, electronic ear tags); - animal location and theft prevention (GNSS); - grazing management (remote sensing); - animal behaviour, activity time budgets, grazing intake (ACC, GNSS, IMU); - drinking, feeding (ACC, IMU); - virtual fencing; - reproduction monitoring.
Mao et al., 2023	Systematic, keyword-based on Google Scholar and Scopus databases	Automated animal activity recognition algorithm, i.e. on board deep learning algorithm. Indications of existing commercial sensors.

References	Review methodology	General output
Hlimi et al., 2024	Systematic literature review was based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines	PLF to characterise animal behaviour. Classification of wearable devices and related methods to extract behaviour (GNSS, ACC, or combined). Classification of non-wearable (video observations with fixed cameras or UAV). Use of smartphones and smartphone apps. Virtual fencing.
Ding et al., 2025	Systematic, keyword-based Web of Science database review (years of reference 2014-2024)	Review of behaviour monitoring, classifying: -sensor type (GNSS, ACC, but also GYR, PRS, acoustics) -single sensor vs multi-sensor studies -sensor deployment position -processes and algorithms for behavioural recognition.
Lamanna et al., 2025	Systematic, keywords-based review of Scopus, Web of Science, and Google Scholar	Sensors integrated into collar-based systems, detailing their functionalities and applications. It underlines the variety (ACC, RFID tags, GNSS, microphones, GYR, MAG), the opportunities (live tracking, little invasive, behavioural analysis) and the challenges (energy consumption, limited parametrization). Underlines the need to integrate multiple sensors.
Bernabucci et al., 2025	Systematic, keyword-based Scopus database review. AND Perspective evaluation of main technologies.	1) Temporal evolution and geographical distribution of PLEF literature over years (main topics, gaps. 2) PLEF sensors, noticing that: - PLEF shifts livestock system management from manual to (semi-) automated. - Text mining shows the evolution and spread of PLEF literature over the years. - PLEF tools help address environmental and livestock system challenges. 3) Classification of outputs of PLEF applied to animal management (animal location and tracking; animal identification; body weight and body condition; behaviour monitoring; feeding behaviour; Oestrus detection; parturition event; pedigree determination; environmental impact monitoring; grazing intensity; transmission of disease; predation of livestock).

Source: compiled by the author based on a mixed literature search methodology.

Table 7. Main biologging sensors and technology for livestock identification and tracking (mainly PLEF).

Technology	Information	Description	References
GNSS collars [Loggers and Sensors]	Identification, detection, location, movement, space use, grazing distribution.	Most deployments of GNSS (usually GPS) collars on livestock (mainly cows, more rarely sheep/goats) have been experimental assessments of grazing distribution and efficiency. Recently, some studies have aimed at extending the data gathering at the seasonal level, typically corresponding with the time spent by livestock in rangeland. In case of GNSS collars, transmission based on GSM or satellites. Further applications consider miniature GNSS units of promising application, although clearly paying in terms of duration.	Brennan et al., 2021 McGranahan et al., 2018 Zhang et al., 2023 Knight et al., 2018 Stewart et al., 2025
GNSS tags [Loggers and Sensors]	Identification, detection, location, movement, space use, grazing distribution.	The miniaturization of GNSS units and the optimisation of energy use in the phase of data transmission (Low Power GNSS units) has allowed the experimentation and the scalability of GNSS tags, that also exists in commercial products. Given the small dimension and reduced battery capability, GNSS tags mostly rely on low power transmission technologies (e.g. LPWAN), yet the utilization of dedicated constellations of small satellites represents a frontier for the utilization over large scales.	Hofman et al., 2022 Moore et al., 2023 Stewart et al., 2025
ACC [Loggers and Sensors]	Behaviour, energy expenditure, energy budget.	Accelerometers have a relatively long history in PLF, as they have a huge value for the evaluation of behaviours, hence energy expenditure and grazing profitability. Accelerometers are deployed on both cattle and sheep/goats (more rarely on other species), often using tags. Experimental deployments have considered attachments also on hind legs in sheep and heels for cows.	Bailey et al., 2008 Ungar et al., 2005 Simanungkalit et al., 2021
ACC & GNSS IMU & GNSS [Loggers and Sensors]	Identification, detection, behaviour, energy expenditure, energy budget, grazing efficiency, feeding behaviour, health status.	Accelerometers provide a full understanding of the linkage between livestock behaviour and grazing land when combined with GNSS. Given the higher energy demand, the combination requires larger units, relying on collars. The potential for the combination of location sensors and accelerometers is unique to derive movements on rangeland behaviors of farming animals (see below), so measuring their grazing efficiency, or predicting their health status. The combination of GNSS and ACC brings PLEF closer to the opportunities offered by the monitoring of housed animals. The downside is the cost, the analytical challenges, and an unlikely scalability. ACC can also be combined with other motion sensors, such as MAG and GYR, all over named Inertia Measurement Units (IMU), allowing a detailed analysis of behaviour and potentially the trajectory reconstruction of animals on rangeland. The downside is the analytical demanding phase and an unlikely scalability, also for difficult data retrieval (but see below LPWAN systems).	Bayley et al 2018 Augustine and Derner, 2013 Anderson et al., 2013 Casas et al., 2021

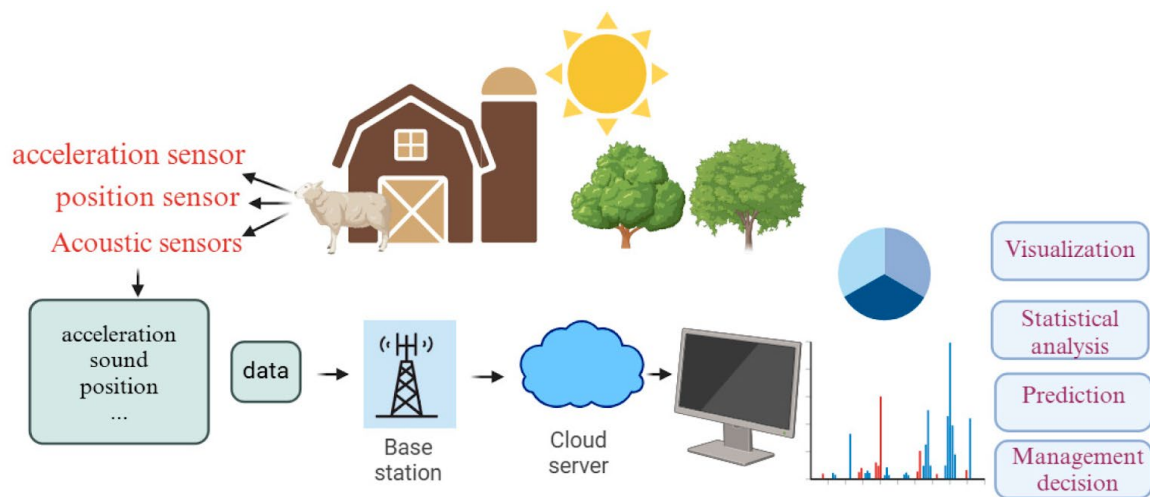
Technology	Information	Description	References
Multi-sensor [Loggers and Sensors]	All the above and physiological measures.	More complex units and microcomputers nowadays easily accommodate additional sensors, such as temperature, light etc. that can provide additional understanding of the external context experienced by farming animals. While these sensors are common for housed animals (for example to monitor welfare conditions of barns or identify sickness), they are still underutilized in PLEF. They do not imply much more energy consumption than combination GNSSxACC. They are typically included in units deployed with collars. Other types of sensors of this kind can be experimentally applied for specific reasons, for example to jaws to monitor bite frequency- yet these applications are more common for housed animals.	Painter et al., 2024 Ding et al., 2025 Arcidiacono et al., 2021 Di Virgilio et al., 2018
Ruminal bolus [Loggers and Sensors]	Identification and Physiological measures.	Although ruminal bolus units do not provide tracking, they offer identification and can be associated with other external tags to obtain important information on metabolism and health status (temperature, heartbeat).	Lee and Seo, 2021
IOT (SigFox) x GNSS [Transmission and 2-way communication]	Location acquisition and transmission.	Locations obtained with GNSS constellations can be transferred to the LPWAN IoT servers, such as SigFox or LoRa, and then to mobile devices easily accessible to shepherds (Figure 3). These networks have a continental cover, they are inexpensive and when used highly increase the cost-effectiveness and energy efficiency of GNSS units, which could be housed also on ear tags. The downside is gaps in IoT networks in remote areas, where typically extensive farming occurs, and an uneven transmission in areas with complex topography, again typical of extensive farming.	Parlato et al., 2024 Mancuso et al., 2023 Vidal et al., 2025
IoT LPWAN - other [Transmission and 2-way communication]	Location acquisition and transmission.	The uneven occurrence of IoT networks in extensive farming areas could be compensated with an additional radio-transmission system, operating with base-stations whose frequency and energy consumption is optimized to the purpose and can even support a continuous flow of IMU data. This system is proposed at hoc and not yet scaled up (Figure 4).	Casas et al., 2021
Assessment of virtual fences/ Classification of behaviours [sensors/data processing]	Grazing management, prevention of theft, behaviour.	GNSS units can be programmed to respond to virtual fences, i.e. to indicate when rangeland-farmed animals are leaving a certain area going beyond the farm property or entering a dangerous area. To this end, virtual fences can therefore be used to monitor grazing distribution, to protect the herd from dangerous areas or from theft. Some systems 'condition' animals to remain within the virtual fence with associated audio systems. While operating well in general, the risk is a discontinuity of service e.g. for data transmission gaps. They typically rely on private servers/systems for data management.	Hassan-Vásquez et al., 2022 Lund et al., 2024 Versluijs et al., 2023

Technology	Information	Description	References
Classification of behaviours (mainly from ACC/GNSS or IMU/GNSS) [data processing]	Behaviour	The exceptional potential of the combination GNSSxACC relies on complex calibration and analytical frameworks to derive behaviours from acceleration data. Despite the complexity, the great potential of these outputs makes this an extremely active area of research. The main goal of current research is to use on edge AI to filter accelerometry data and classify behaviours, so optimizing energy and memory use of the sensory units. There is not yet a scaled-up system, but the enormous progress of AI makes the goal realistic. Ironically, the speed of such development could be much faster in PLEF than in wildlife biologging: the key obstacle for classification of behaviours is the calibration with ground-truth observations which is a limiting factor for wild animals, but not domestic ones.	Hassan-Vásquez et al., 2022 Campiotti et al., 2023 Walton et al., 2018 Kaler et al., 2020 Jin et al., 2022 Jin et al., 2024 Wang et al., 2023
Animal welfare/ Accuracy of collars [other]	Sensor efficiency	The accuracy of sensors, their energy efficiency, and the effects of deployment on animal welfare is not so often taken into consideration or made a priority because of two apparently contradictory reasons: (i) solutions are still in the engineering phase, and not taken to scaled deployments; (ii) or products are off-the-shelf and assumed tested and calibrated- which is not often the case for biologging in general.	Herlin et al., 2021 Schoenecker et al., 2024
UAV [loggers and sensors]	Herd distribution, grazing distribution, Identification	A new frontier to monitor the distribution of herds or individuals within the herd is the utilization of cameras mounted on UAVs. On edge AI systems may in the future also allow the identification of single individuals. This system does not allow continuous monitoring, but if combined with other tagging methods it may have high potential, also considering these systems have a very rapid development for many other civil needs.	Barbedo et al., 2020 Aburasain et al., 2021 Shao et al., 2020 Rivas et al., 2018
Database development [sensors/data processing]	na	The management of data is a challenge of PLEF for data standardization, persistence, trustfulness and security. This aspect is under-considered (see Chapter 4).	Guarda-Vera et al., 2025 Di Virgilio et al., 2018

Source: compiled by the author.

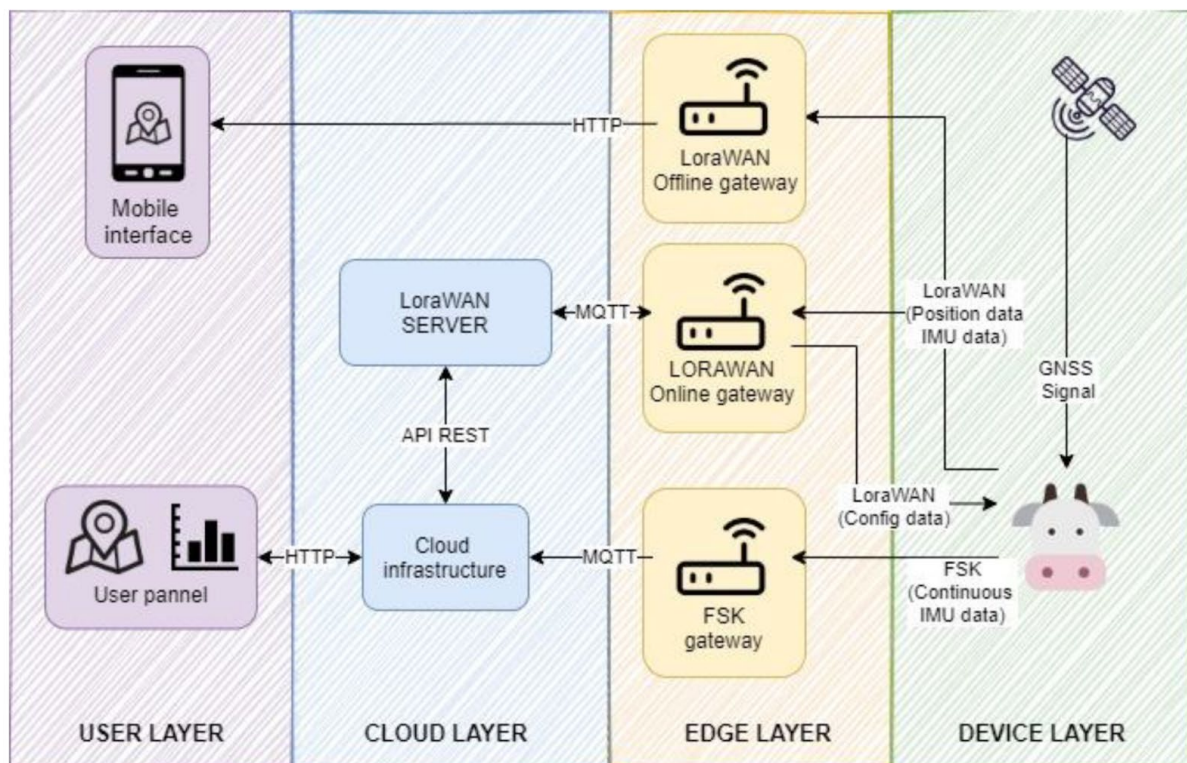
Note: The description of the technology refers to the categories identified in the requirement analysis, **Table 4**.

Figure 3. Example of biologging network deployed on livestock and farming systems (either PLF or PEFL).



Source: Ding et al., 2025.

Figure 4. Example of communication pathways from PEFL biologists to end users.



Source: Casas et al., 2021.

One relevant aspect emerging from the cross-check between published literature (including reviews) and commercial web search, is the **availability of off-the-shelf market products** that aim at the **deployment of light, energy efficient, multi-sensor tags at scale** (Figure 5; Table 8). While some of these units are still experimental (Figure 5 a-c), others have been already deployed over a very large scale (Figure 5 d-g).

Figure 5. Examples of smart ear and leg tags deployed on livestock. From top left: cows wearing a collar (a) and an ear tag (b) used for data collection on behaviour. Sheep wearing Inertial Measurement Unit sensors (c). CERES TAG, a commercial solar-powered satellite tracking device (d) developed for rangeland animals, and an application on the field (e). MOOvement, a commercial solar-powered tag relying on LPWAN developed for cows (f), and an application in the field (g).



Source: (a,b) Wang et al., 2023; (c) Jin et al. 2024; (d,e) <https://cerestag.com/>; (f,g) <https://www.moovement.com.au/>.

Table 8. Available PLEF market products as per a systematic review.

Name	Species	Sensors	Outputs	Country	Website
Pigwei®, Ymaging	Pig	Imagine analysis	Live weight	Spain	http://www.ymaging.com/projects-2/
Datamuster®	Cattle	Walk-over-weigh (weighing crate)	Maternal parentage, reproductive efficiency, growth rates, calving, property mapping	Australia	https://www.datamuster.net.au/
smaXtec®, GmbH	Cattle	Accelerometer, thermometer	pH, body temperature, calving, heat, health, rumination	Austria, Germany	https://smaxtec.com/en/
Ceres Tag®, CeresTag Pty Ltd	Cattle	GNSS, Accelerometer	Activity, Geofencing, health	Australia	https://www.cerestag.com/
digitanimal®	Cattle, horses, sheep, goats	GNSS, Accelerometer, thermometer	Activity, Geofencing, body temperature	Spain	https://digitanimal.com/?lang=en
Allflex SenseHub®, SCR Engineers Ltd.	Cattle	Accelerometer	Health, rumination, intake, heat, calving, activity, heat stress	Israel	https://www.allflex.global/
Moomonitor+®, Dairymaster	Cattle	Accelerometer	Activity, resting, feeding, rumination, heat detection	Ireland	https://www.dairymaster.com/
IceTag® and IceQube®, IceRobotics, Ltd	Cattle	Accelerometer	Lameness, activity, resting, heat detection	UK	https://www.icerobotics.com/
Moocall®	Cattle	Accelerometer	Calving, heat detection	Ireland	https://www.moocall.com/
CalveSense®, SCR Engineers Ltd.	Cattle	Accelerometer	Calving	Israel	https://www.allflex.global/

Name	Species	Sensors	Outputs	Country	Website
eShepherd®, Agersens	Cattle	GNSS, accelerometer	Virtual fencing, activity monitoring, pasture management	Australia	https://agersens.com/
Halter®	Cattle	GNSS, accelerometer	Virtual fencing, activity monitoring, pasture management	New Zealand	https://halterhq.com/
Vence®	Cattle	GNSS, accelerometer	Virtual fencing, activity monitoring, pasture management	USA	http://vence.io/
Nofence®	Cattle, sheep, goats	GNSS, accelerometer	Virtual fencing, activity monitoring	Norway	https://www.nofence.no/en

Source: Aquilani et al., 2022.

3.5. Scaled-up biologging applications to extensive or hybrid farming: data management and preservation caveats

While the diversity of sensors and products derived from the literature review on PLEF is promising for a fast development in the field that could also **match the EU requirements** (Chapter 2), some aspects are more hidden **and potentially hindering the process**.

- **Data standards:** there is very little available on the standards to be used for the data obtained from the sensors, and even less on the calibration and validation of tools, a critical aspect also identified in the wildlife biologging context (Payne et al., 2026).
- Similarly, **the data flow**, from tags to repositories, is not explicit, mostly closed, and with unclear statements on **data accessibility, ownership, and legal preservation**— a key aspect for livestock. This could be a **key limitation for the utilization of off-the-shelf commercial tags** in the EU. Specific assessments would be needed to evaluate the compatibility with the General Data Protection Regulation (GDPR, EU Regulations 2016/679).

This points at a crucial aspect: while data are pushed to be open when linked to research, information on livestock shall be traceable, reliable, accessible and interoperable between systems, but it is also **subject to confidentiality, privacy, top-down (i.e. not open management) in critical cases, for example in case of disease spread**.

4. Feasibility and scalability of biologging applications for tracking livestock movement in the EU policy context.

4.1. Potential technical solutions for tracking livestock movements and behaviour vs EU policy requirements

From a purely **technical perspective**, matching **Table 3** (legal requirements for farming animal identification and tracking) with what obtained by the literature review on PLEF (**Table 6, Table 7**), while considering the general requirements for biologging deployment (**Table 4**), knowledge derived from wildlife biologging (**Table 5**), and always bearing in mind the **objectives** seek for by the Strategic Dialogue on the future of EU agriculture (One Health, Animal Welfare, Innovation and Reward, **Table 2**), leads to some **technical options for the integration of sensors for tracking livestock movements and behaviour with the electronic identifiers (e.g. tags) already required/permitted by EU policies**, as indicated in **Table 9**.

Table 9. Potential use of biologging sensors associated with external electronic identifiers (ear tags and pastern bands) for livestock (extensive/hybrid: PLEF).

External electronic identifier	Bovine	Ovine <12 months	Ovine >12 months	Porcine	Equine	Camelid	Cervid
Electronic ear tag	GNSS eartags, IoT eartags, other sensors	NA	GNSS eartags, IoT eartags	GNSS eartags, IoT eartags, other sensors	NA§	NA§	NA§
Electronic pastern band	NA	NA	GNSS pastern band, IoT pastern band	NA	NA	NA	NA
Ruminal Bolus	Temperature, heartbeat sensors	NA	NA	NA	NA	NA	NA

§ In exceptional cases the use of an ear tag for Equine is possible; Electronic tags are not mentioned in the Delegated Regulation for Equine, Camelid and Cervid. If the possibility is positively discussed, the same choice as per Bovine is suggested, i.e. GNSS eartags, IoT eartags, other sensors

Source: compiled by the author.

The essential aspects to be considered are:

- the choice of **electronic ear tags offering localization beyond RFID for identification as an optimal compromise between technological possibilities and legislation requirements**, with localization obtained either by GNSS or IoT triangulation systems as a choice of reference (Objective 1: One Health, Objective 3: Innovation);

- the consideration of transmission methods as a key aspect for the evaluation of the effective and scalable use of these technologies (**Table 4, Figure 1**), with retrieval from either satellite systems or regional IoT networks as a choice of reference (Objective 3: Innovation, Objective 4: Reward);
- the consideration that the combination of GNSS and accelerometry to extract behavioural information would be extremely valuable to increase the sustainability and quality of production, including animal welfare (Objective 1: One Health, Objective 2: Animal Welfare), but it is still difficult to scale up, given the need of calibration and sophisticated analytical tools to extract behavioural information from such data, so indicating additional sensors to localization as optional;
- that additional sensors complementing identification and localisation with physiological parameters should be potentially integrated (like rumen bolus for bovines; Objective 1: One Health, Objective 2 Animal welfare).

Finally, considering biologgers as a viable solution for domestic animals should also consider the further limitations imposed by the aesthetic value of individuals (e.g. horses), or the need to apply scalable approaches (e.g. camelid, cervids).

4.2. Challenges for upscaling technical solutions at EU level: data transmission, welfare, cost

Beyond identifying the potential technical solutions for livestock electronic tagging based on availability of sensors and current policy requirements (**Table 9**), it is necessary to operate further considerations regarding the **scalability of such solutions at the EU level** (Strategic Dialogue on the future of EU agriculture, EU Commission, 2024).

The following elements are not exhaustive limitations, but they would merit **particular attention and further assessment** before uptaking specific solutions.

1. **Data transmission and preservation.** What mentioned in Chapter 3.5 represents important technical caveats and limitations that may prevent the scalability of electronic tagging solutions at the EU level. The main technical limitations do not currently sit in the core technology (data acquisition), but in the successive stages of the data life, i.e. data transmission and preservation (**Table 4**). Some key aspects to consider are:
 - a. How **efficiently** are the data transmitted? What is the likelihood for data **transmission gaps**? What is the **coverage** for data transmission? What is the **cost** for data transmission?
 - b. **Where are the data transmitted to? Is the server proprietary?** Is it safe?
 - c. **Where are the data preserved**, in what database? Does it comply with privacy and sensitivity of the managed information, and GDPR? Can it be **integrated with IACS/CAP information flow at the national level** (Jänicke et al., 2025)?

Some potential opportunities and perspectives are discussed in Chapter 5, having in mind the **Objective 3: Innovation and Objective 4: Reward** of the Strategic Dialogue on the future of EU agriculture (EU Commission, 2024).

2. **Electronic tagging packaging and animal welfare.** The needed electronics for localization and data transmission should ‘fit’ into an ear tag that complies with animal welfare requirements (**Objective 2- Animal welfare**), for example not altering the ear shape, and being of minimal weight. That said, **digitalisation** would in general facilitate the understanding of animal welfare status with respect to traditional methods (**Objective 3- Innovation**). However, the potential impact of new tracking technologies on animal welfare is still under evaluation (Tuytens et al., 2022, Payne et al., 2026) and is not explored in detail in this report.
3. **Utility and ease of uptake from the community of stakeholders** (farming community, pastoralism). Following the principles expressed at the Strategic Dialogue on the future of EU agriculture, the technological advancements when scaled up must improve the quality of life of the farming community, provide **reward for sustainable practices**, while **decreasing the administrative load (Objective 3: Innovation; Objective 4: Reward)**. A technically perfect solutions that is over-complicated and requires too much training would therefore not be an optimal one. Some potential opportunities and perspectives are discussed in Chapter 5.
4. **Cost.** Electronic units for wildlife species are currently quite expensive- in the order of €1 000-€3 000/unit for middle-sized ungulates for GNSS collars. The unit cost of alternative solutions, though, such as **IoT collars or potentially ear tags is much lower (in the order of €100-€300/unit)**. The unit cost could also be reduced with mass production of commercial products, for example by Tenders for R&T innovation for industry and SMEs (Chapter 5). The cost for data transmission should also be added, which however could be optimised with centralised solutions harmonised with EU remote sensing initiatives (Chapter 5). Finally, if the actual costs for the implementation of these technologies will require a relevant financial effort by farmers, it might also be supported by incentives (e.g. CAP support).

5. Opportunities and perspectives for the implementation of smarter farming (smart pastures)

5.1. Possible approaches for turning technological opportunities into pragmatic solutions

The innovation presented throughout the review and revisited against the EU policy context in Chapter 4.1 is a concrete possibility for a PLEF at the EU level, considering the innovative solutions for wildlife species (Chapter 3.3) as promising pathways. Miniaturization and multi-sensoring for the electronic tags are a key perspective to consider beyond the basic requirement of identifying and tracking animals, with the potential of closing the gap between PLF and PLEF in terms of data comprehensiveness on extensive farming productivity, to promote sustainability and quality of production, agroecology, agroforestry, regenerative agriculture, and pastoralism. Electronic tags with localization sensors (GNSS) and occasionally other sensors **may become the reference identification system**, but a series of **actions for scalability and uptake by the farming community** could be considered to address the challenges identified in Chapter 4.2.

- **Data transmission and preservation, and cost. A dedicated, EU-based system of micro-satellites for data transmission and electronic tag monitoring could be put in place.** This would ensure retrieval of GNSS localization and control on the tags, while also assuring EU GDPR and privacy protocols are respected. To this end, collaboration with Copernicus could be considered. Such infrastructure for data transmission and retrieval could be sided by a seamless data flow in dynamic databases such as TRACES, providing information at Member States level for the implementation of IACS to manage CAP support. As data transmission would otherwise be a live cost for the farming community, this approach would also reduce the overall e-tag unit cost for the final users and facilitate the farming community uptake. Connecting with IACS would contribute to the CAP objectives, improving the assessment of low-intensity grazing systems, confirming active grazing, and validating the remote sensing-based models (see Chapter 2.2). This would contribute to the general CAP goals of simplification and reduction of administrative burden for farmers, increase digitisation in agriculture and rural areas, and generational renewal. Regional IoT infrastructures could also support data collection and animal monitoring and allow the use of IoT-based triangulation as an alternative to GNSS, complementing a purely GNSS/data transmission satellites system. IoT infrastructure would have a deployment and maintenance cost, but they could be used for many other data flows. Existing private networks (LoRaWAN, SigFox) could be leveraged if in compliance with EU regulations.
- **Electronic tagging packaging, animal welfare and cost.** Once the data transmission and preservation challenge is addressed, the production at scale of electronic tags for PLEF which are compliant with the aforementioned requirements and respect animal welfare could be operated by private partners (e.g. through a Tender procedure), so also prompting competitiveness and innovation of EU technological industry. This mechanism would promote mass production and reduction of unit cost of electronic tags for individual identification and tracking, making them affordable and cost-effective with respect to the advantages (better production, easier control, see below).

- **Utility and ease of uptake from the community of stakeholders** (farming community, pastoralism). As the suggested solutions are e-tags, similar to what already used for livestock, but technologically more advanced, not much would change in the identification procedures for farmers. Conversely, if a centralised system for data transmission and retrieval is developed (such as microsatellites for data retrieval, see above), this would ultimately reduce the administrative load (animals would operate a ‘self check-in’ in the traceability systems) and promote sustainability. Also, this would support the transition ‘*from compliance to performance*’ also for extensive farming, as already operated by CAP 2023-2027 for the Area Monitoring System. It would also improve the quality of life of the extensive farming community, by improving the competitiveness of the products (via transparency of the production workflow and the reliability of products labelling) and more viable life-work balance (by remote monitoring). However, these systems should be tested and evaluated for the resilience to bugs, hacking, power cuts, and for the ease of use. Ideally, simple mobile phone applications should allow the farmer to monitor and transmit the information about their farmed animals, as already happens for other type of information. As these aspects need thorough testing, pilot studies could be run in the context of existing initiatives and funded projects (see Chapter 5.2).

5.2. Opportunities for coordination with existing initiatives

Several of the available technologies and especially the more innovative and still experimental ones need factual testing and evaluation for their scalability. Leveraging existing initiatives, groups of interest and EU-funded projects is a fundamental step to (i) increase the potential for operationalise innovation; (ii) decrease the timelag between proposed innovation and scalable use, while operating with caution to avoid mass failures; (iii) optimise public resources and shared knowledge.

To this end, **Table 10** offers an overview of the most pertinent initiatives and projects that are just started, ongoing, or recently ended. Several opportunities emerge, with work packages or initiatives aiming to test innovative solutions to detect, track and monitor with biologging.

Table 10. Potential for leveraging existing projects and initiatives for introducing biologging for farmed animals.

Initiatives	Description	Opportunities
Agriculture of Data (AgData) ¹ (2025-32)	AgData will enhance the use of data for developing data-based solutions, methodologies, and indicators to improve climate adaptation and resilience of agriculture as well as to minimize the undesired impact of agriculture on the climate, environmental quality, and ecosystem services.	Use Cases (UCs), the AgData partnership will deploy a broad variety of activities in all relevant branches of agriculture based on modern digital-technologies and data sources. The UCs will all be based on the Strategic Research and Innovation Agenda (SRIA). Besides the UCs AgData will also implement joint calls for transnational proposals resulting in financial support to third parties (FSTP). As the project has just started, UC could be an opportunity to test biologging tools for farm animals as proposed in Chapter 3.

Initiatives	Description	Opportunities
DigiRangeland ² (2025-29)	Rangeland livestock farming systems (RLFS) are integral to sustainable development of agriculture in Europe, supporting the European Green Deal through vital ecosystem services. However, these systems face numerous challenges that threaten their viability. The EU-funded DIGI-Rangeland project seeks to address these challenges by creating an innovative network focused on digital technologies and data solutions tailored for farmers and land users in rangeland areas.	The project aims also at identifying and evaluating solutions that align with the needs of RLFS farmers and land users, also considering the needs, obstacles and expectations regarding digital technologies. As the project has just started, it could represent an opportunity to introduce/test/assess innovation on e-tagging of livestock as proposed in the Chapter 3.
Digi4Live ³ (2024-28)	Six case studies. These case studies will focus on solutions for monitoring animals, their surroundings, and the environmental effects. Each study tackles both practical and technical challenges with the goal of being scalable across the European Union. See for example case study 'Robust AI methods for sensor based animal tracking' https://digi4live.eu/cs-c-robust-ai-methods-for-sensor-based-animal-tracking/ . Specifically, it uses computer vision for pigs and telemetry data for cows (animal tracking and welfare assessment), with the objective of embedded sensors and computer vision, enhanced by AI, enabling automated tracking of animal locations and behaviours. Also seen in the context of smart pastures: by using IoT, remote monitoring, and AI, smart pastures optimize grazing, enhance productivity and reduce the environmental footprint. In CS-C, the integration of IoT sensors and advanced AI algorithms ensures precise tracking and analysis of livestock and pasture conditions. These solutions optimize pasture use and improve pasture management, contributing to more productive and sustainable farming systems. Smart pastures vs precision husbandry.	Ensuring the robustness of AI algorithms in tracking animal access to outdoor environments and generating animal welfare Key Performance Indicators (KPIs) requires rigorous testing in diverse environmental conditions before widespread adoption of these solutions. Smart pastures leverage digital technologies to transform traditional livestock farming into more sustainable and data-driven practices- Internet of Cows, remote sensing (drones), AI to process output data. All these solutions have been envisioned as perspective for innovation- hence the project could be an opportunity to see them operationalised. Yet, several of the proposed approaches do not comply with the requirements presented in Chapter 2 and Chapter 3.

Initiatives	Description	Opportunities
LIFE PROJECTS (1): LIFE LiveAdapt ⁴ (concluded) LIFEstockProtect ⁵ (2020-2026) See also: CAP2ER ⁶	In general, these projects draw attention to climate adaptation, protection from predators; there is also a strong attention on the effect of agriculture on biodiversity and soil.	The opportunities for livestock tracking expressed in Chapter 2 and in Chapter 3 are well harmonised with these needs. Any monitoring offered in the context of the CAP shall also help in this sense.
LIFE PROJECTS (2): PASTORALP ⁷ (concluded) SHEPFORBIO ⁸ (2021-27)	Pastures vulnerability and adaptation strategies to climate change impacts in the Alps; and Shepherds for Biodiversity in Mountain Marginal Areas.	The latter project is ongoing and it could represent an opportunity to meet stakeholders and propose alternative monitoring means for livestock in less-served areas.
NoFence ⁹	Norwegian technology company providing virtual fencing for livestock (cattle, sheep, goats) using GNSS collars and a smartphone app to define digital boundaries, eliminating the need for physical fences and allowing flexible grazing management.	The principle is very useful for hybrid and extensive farming, although at the moment limited to collars. Implementation with ear tags would have great potential when reconciled with the requirements mentioned in Chapter 2 and Chapter 3.
Vistock ¹⁰	NoFence technology for farms in Tuscany.	Lessons learned from the deployment.
Euromontana ¹¹	The multi-stakeholder network Euromontana supports pastoralism in terms of quality of life, advocating for greater support, focusing on working conditions, promotion of products and constructive dialogue. They strive to adopt a European Action Plan on Pastoralism; and a 'Mountain product' quality brand.	Taking advantage of the discussion on the CAP reform post-2027, Euromontana proposes to raise awareness on pastoralism in Europe, using CAP to support pastoral activities. It could therefore be a perfect reference to assess and co-develop solutions for better digital tracking of livestock in hybrid/extensive and pastoral systems.
IDELE ¹²	French Livestock Institute (IDELE) is the French national technical reference and normative body in livestock farming systems. It is a non-profit, non-governmental organization featuring management by livestock farmer's federations and trade unions.	They can prototype, test and validate digital solutions, offering a great opportunity to test innovation in e-tags dedicated to livestock in extensive pastures and systems.

Source: compiled by the author (see notes for references to the websites used to collect information).

Notes: ¹ <https://cordis.europa.eu/project/id/101197332>; ² <https://cordis.europa.eu/project/id/101183132>; ³ <https://digi4live.eu/project-overview-2/>; ⁴ <https://climate-adapt.eea.europa.eu/en/metadata/projects/adaptation-to-climate-change-of-extensive-livestock-production-models-in-europe-1/>; ⁵ <https://lifestockprotect.info/en/project-overview/project-details/>; ⁶ <https://digiagrifood.ch/it/digiblogs/cap2er-calcolo-delle-prestazioni-ambientali-delle-aziende-agricole/>; ⁷ <https://www.pastoralp.eu/home/>; ⁸ <https://dream-italia-euprj.eu/en/life-shepforbio/>; ⁹ <https://www.nofence.com/>; ¹⁰ <https://vistock.toscanallevatori.it/>, <https://www.sciencedirect.com/science/article/abs/pii/S1161030122001952>; ¹¹ <https://www.euromontana.org/wt-pastoralism/>; ¹² <https://idele.fr/>, <https://european-digital-innovation-hubs.ec.europa.eu/edih-catalogue/idele>.

6. Conclusions

The evaluation of biologging technologies for identification and monitoring of livestock in the EU, with special reference to extensive or hybrid farming and pastoralism leads to several key conclusions regarding their potential viability and implementation.

- **Technological readiness and transition:** Biologging is a mature field with 80 years of scientific research that is now ready to transition from biodiversity monitoring to **Precision Livestock Extensive Farming (PLEF)**.
- **Optimal technical solutions:** Integrating **localization sensors (GNSS or IoT)** into authorized **electronic ear tags** (Commission Delegated Regulation (EU) 2019/2035) represents the best compromise between current EU legal requirements and technological potential.
- **Strategic alignment with EU policy:** These innovations directly support the **Strategic Dialogue on the future of EU agriculture** by promoting the **OneHealth** approach, enhancing **animal welfare**, boosting **innovation** and **rewarding** sustainable practices like extensive farming and pastoralism.
- **Administrative simplification:** Implementing live-tracking allows for **‘self-checking’ animals**, which significantly reduces the administrative burden on farmers and moves the livestock system from a model of **compliance to performance**, following the CAP 2023-2027 approach implanted with the AMS for the monitoring of area-based interventions, and evaluated for extension to the farming systems post-2027.
- **Data preservation and security/Infrastructure:** A major hurdle for scalability is the lack of explicit **data standards** and the use of **proprietary servers**, which may hinder interoperability and compliance with **GDPR**, but this could be resolved by a **centralized EU-based system of micro-satellites** for data transmission (potentially, in the framework of Copernicus).
- **Economic scalability and unit cost:** The cost of single biologging units should be minimised to allow accessibility by farmers. This could be obtained by **mass production** of tags via private-public partnerships, and inclusion of these innovative identification approaches into CAP support system to farmers. The actual financial scalability should be subject of verification.
- **Smart pastures, happy pastures:** The opportunity for a more automated verification of localisation of spatial distribution of livestock, and the related grazing load, can concretely improve the quality of life of farmers, with particular reference to extensive farming and pastoralism. It also concretely contributes to biodiversity, by providing the instruments to avoid grazing overload and nitrogen unbalance.
- **Future implementation via pilot programs:** Leveraging existing initiatives and funded projects like **AgData** or **Digi4Live** is essential to test these technologies for resilience against technical failures and to ensure they are user-friendly for and accepted by the farming community.

References

- Aburasain, R. Y., Edirisinghe, E. A., & Albatay, A. (2021). Drone-based cattle detection using deep neural networks. In K. Arai, S. Kapoor, & R. Bhatia (Eds.), *Intelligent Systems and Applications* (pp. 598–611). Springer. https://doi.org/10.1007/978-3-030-55180-3_44
- AlZahal, O., AlZahal, H., Steele, M. A., Van Schaik, M., Kyriazakis, I., Duffield, T. F., & McBride, B. W. (2011). The use of a radiotelemetric ruminal bolus to detect body temperature changes in lactating dairy cattle. *Journal of Dairy Science*, 94(7), 3568–3574. <https://doi.org/10.3168/jds.2010-4067>
- Ambriz-Vilchis, V., Jessop, N. S., Fawcett, R. H., Shaw, D. J., & Macrae, A. I. (2015). Comparison of rumination activity measured using rumination collars against direct visual observations and analysis of video recordings of dairy cows in commercial farm environments. *Journal of Dairy Science*, 98(3), 1750–1758. <https://doi.org/10.3168/jds.2014-8538>
- Anderson, D. M., Estell, R. E., Holechek, J. L., Ivey, S., & Smith, G. B. (2014). Virtual herding for flexible livestock management – a review. *The Rangeland Journal*, 36(3), 205–221. <https://doi.org/10.1071/RJ13092>
- Aquilani, C., Confessore, A., Bozzi, R., Sirtori, F., & Pugliese, C. (2022). Precision Livestock Farming technologies in pasture-based livestock systems. *Animal*, 16(1), 100429. <https://doi.org/10.1016/j.animal.2021.100429>
- Arcidiacono, C., Mancino, M., Porto, S. M. C., Bloch, V., & Pastell, M. (2021). IoT device-based data acquisition system with on-board computation of variables for cow behaviour recognition. *Computers and Electronics in Agriculture*, 191, 106500. <https://doi.org/10.1016/j.compag.2021.106500>
- Augustine, D. J., & Derner, J. D. (2013). Assessing herbivore foraging behavior with GPS collars in a semiarid grassland. *Sensors*, 13(3), 3711–3723. <https://doi.org/10.3390/s130303711>
- Bailey, D. W., Trotter, M. G., Knight, C. W., & Thomas, M. G. (2018). Use of GPS tracking collars and accelerometers for rangeland livestock production research. *Translational Animal Science*, 2(1), 81–88. <https://doi.org/10.1093/tas/txx006>
- Bailey, D. W., Trotter, M. G., Tobin, C., & Thomas, M. G. (2021). Opportunities to apply precision livestock management on rangelands. *Frontiers in Sustainable Food Systems*, 5, 611915. <https://doi.org/10.3389/fsufs.2021.611915>
- Bailey, D. W., VanWagoner, H. C., Weinmeister, R., & Jensen, D. (2008). Comparison of low-moisture blocks and salt for manipulating grazing patterns of beef cows. *Journal of Animal Science*, 86(5), 1271–1277. <https://doi.org/10.2527/jas.2007-0578>
- Barbedo, J. G. A., Koenigkan, L. V., Santos, P. M., & Ribeiro, A. R. B. (2020). Counting cattle in UAV images—Dealing with clustered animals and animal/background contrast changes. *Sensors*, 20(7), 2126. <https://doi.org/10.3390/s20072126>
- Barbknecht, A. E., Fairbanks, W. S., Rogerson, J. D., Maichak, E. J., & Meadows, L. L. (2009). Effectiveness of vaginal-implant transmitters for locating elk parturition sites. *The Journal of Wildlife Management*, 73(1), 144–148. <https://doi.org/10.2193/2007-428>
- Beardsworth, C. E., Gobbens, E., van Maarseveen, F., Denissen, B., Dekinga, A., Nathan, R., Toledo, S., & Bijleveld, A. I. (2022). Validating ATLAS: A regional-scale high-throughput tracking system. *Methods in Ecology and Evolution*, 13(9), 1990–2004. <https://doi.org/10.1111/2041-210X.13913>

- Berckmans, D. (2017). General introduction to precision livestock farming. *Animal Frontiers*, 7(1), 6–11. <https://doi.org/10.2527/af.2017.0102>
- Bernabucci, G., Evangelista, C., Girotti, P., Viola, P., Spina, R., Ronchi, B., ... & Primi, R. (2025). Precision livestock farming: an overview on the application in extensive systems. *Italian Journal of Animal Science*, 24(1), 859–884. <https://www.tandfonline.com/doi/full/10.1080/1828051X.2025.2480821>
- Bijleveld, A. I., van Maarseveen, F., Denissen, B., Dekinga, A., Penning, E., Ersoy, S., Gupte, P. R., de Monte, L., ten Horn, J., Bom, R. A., Toledo, S., Nathan, R., & Beardsworth, C. E. (2022). WATLAS: High-throughput and real-time tracking of many small birds in the Dutch Wadden Sea. *Animal Biotelemetry*, 10, Article 36. <https://doi.org/10.1186/s40317-022-00307-w>
- Boyce, M. S., & McDonald, L. L. (1999). Relating populations to habitats using resource selection functions. *Trends in ecology & evolution*, 14(7), 268–272. [https://doi.org/10.1016/S0169-5347\(99\)01593-1](https://doi.org/10.1016/S0169-5347(99)01593-1)
- Brennan, J., Johnson, P., & Olson, K. (2021). Classifying season long livestock grazing behavior with the use of a low-cost GPS and accelerometer. *Computers and Electronics in Agriculture*, 181, 105957. <https://doi.org/10.1016/j.compag.2020.105957>
- Broom, D. M. (2019). Animal welfare complementing or conflicting with other sustainability issues. *Applied Animal Behaviour Science*, 219, 104829. <https://doi.org/10.1016/j.applanim.2019.06.010>
- Cagnacci, F., & Massei, G. (2008). Monitoring bait uptake in badgers: A pilot study using PIT-tag system and time-lapse video surveillance. *Ethology Ecology & Evolution*, 20(2), 165–171. <https://doi.org/10.1080/08927014.2008.9522501>
- Cagnacci, F., Boitani, L., Powell, R. A., & Boyce, M. S. (2010). Animal ecology meets GPS-based radiotelemetry: A perfect storm of opportunities and challenges. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 365(1550), 2157–2162. <https://doi.org/10.1098/rstb.2010.0107>
- Campiotti, V., Finozzi, N., Irazoqui, J., Cabrera, V., Ungerfeld, R., & Oreggioni, J. (2023). Wearable device to monitor sheep behavior. 2023 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor), 286–290. <https://doi.org/10.1109/MetroAgriFor57405.2023.10009941>
- Casas, R., Hermosa, A., Marco, A., Blanco, T., & Zarazaga-Soria, F. J. (2021). Real-time extensive livestock monitoring using LPWAN smart wearable and infrastructure. *Applied Sciences*, 11(3), 1240. <https://doi.org/10.3390/app11031240>
- Chimienti, M., Cornulier, T., Owen, E., Bolton, M., Davies, I. M., Travis, J. M. J., & Scott, B. E. (2016). The use of an unsupervised learning approach for characterizing latent behaviors in accelerometer data. *Ecology and Evolution*, 6(3), 727–741. <https://doi.org/10.1002/ece3.1914>
- Corcoran, E., Winsen, M., Sudholz, A., & Hamilton, G. (2021). Automated detection of wildlife using drones: Synthesis, opportunities and constraints. *Methods in Ecology and Evolution*, 12(6), 1103–1114. <https://doi.org/10.1111/2041-210X.13581>
- Caballero García De Arevalo, R., & Van Der Velde, M. (2025). Policy Recommendations for Large-Scale Grazing Systems in Europe: Preserving Territorial Identity and Addressing Pressures. Publications Office of the European Union, Luxembourg. JRC143525. <https://data.europa.eu/doi/10.2760/7528855>

Ding, L., Zhang, C., Yue, Y., Yao, C., Li, Z., Hu, Y., ... & Li, Q. (2025). Wearable sensors-based intelligent sensing and application of animal behaviors: A comprehensive review. *Sensors*, 25(14), 4515. <https://doi.org/10.3390/s25144515>

di Virgilio, A., Morales, J. M., Lambertucci, S. A., Shepard, E. L. C., & Wilson, R. P. (2018). Multi-dimensional precision livestock farming: A potential toolbox for sustainable rangeland management. *PeerJ*, 6, e4867. <https://doi.org/10.7717/peerj.4867>

English, H. M., Börger, L., Kane, A., & Ciuti, S. (2024). Advances in biologging can identify nuanced energetic costs and gains in predators. *Movement Ecology*, 12, Article 7. <https://doi.org/10.1186/s40462-024-00448-y>

European Commission. (2024). A shared perspective for agriculture and food in Europe: Final report of the Strategic Dialogue on the future of EU agriculture. Publications Office of the European Union. https://agriculture.ec.europa.eu/common-agricultural-policy/cap-overview/strategic-dialogue-future-eu-agriculture_en

European Commission. (2025). A vision for agriculture and food (COM/2025/75 final). <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM:2025:75:FIN>

EUROSTAT (2025a). Agricultural production - livestock and meat. Statistics Explained. Retrieved February 10, 2026, from https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Agricultural_production_-_livestock_and_meat

EUROSTAT (2025b). Greenhouse gas emissions from agriculture. Retrieved February 10, 2026 from <https://ec.europa.eu/eurostat/databrowser/view/tai08>

EUROSTAT (2026). Performance of the agricultural sector. Statistics Explained. Retrieved February 10, 2026 from https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Performance_of_the_agricultural_sector

Guarda-Vera, M., & Muñoz-Poblete, C. (2025). Preliminary development of a database for detecting active mounting behaviors using signals acquired from IoT collars in free-grazing cattle. *Sensors*, 25(10), 3233. <https://doi.org/10.3390/s25103233>

Hassan-Vásquez, J. A., Maroto-Molina, F., & Guerrero-Ginel, J. E. (2022). GPS tracking to monitor the spatiotemporal dynamics of cattle behavior and their relationship with feces distribution. *Animals*, 12(18), 2383. <https://doi.org/10.3390/ani12182383>

Herlin, A., Brunberg, E., Hultgren, J., Högberg, N., Rydberg, A., & Skarin, A. (2021). Animal welfare implications of digital tools for monitoring and management of cattle and sheep on pasture. *Animals*, 11(3), 829. <https://doi.org/10.3390/ani11030829>

Hlimi, A., El Otmani, S., Elame, F., Chentouf, M., El Halimi, R., & Chebli, Y. (2024). Application of precision technologies to characterize animal behavior: a review. *Animals*, 14(3), 416. <https://www.mdpi.com/2076-2615/14/3/416>

Hofmann, W., Neal, M., Woodward, S., & O'Neill, T. (2022). GPS technology as a tool to aid pasture management on dairy farms. *Journal of New Zealand Grasslands*, 189-196. 10.33584/jnztg.2022.84.3561

Holbrook, S. (2012, March 13). Farming communities face crisis over nitrate pollution, study says. Food & Environment Reporting Network. <https://thefern.org/2012/03/farming-communities-facing-crisis-over-nitrate-pollution-study-says/>

Jänicke, C., Petersen, K. A., Schmidts, P., Müller, D., & Jepsen, M. R. (2025). Field and farm-level data on agricultural land use for the European Union. *Scientific Data*, 12(1), 1050.

<https://www.nature.com/articles/s41597-025-05210-6>

Jin, Z., Guo, L., Shu, H., Qi, J., Li, Y., Xu, B., Zhang, W., Wang, K., & Wang, W. (2022). Behavior classification and analysis of grazing sheep on pasture with different sward surface heights using machine learning. *Animals*, 12(14), 1744. <https://doi.org/10.3390/ani12141744>

Jin, Z., Shu, H., Hu, T., Jiang, C., Yan, R., Qi, J., ... & Guo, L. (2024). Behavior classification and spatiotemporal analysis of grazing sheep using deep learning. *Computers and Electronics in Agriculture*, 220, 108894. <https://doi.org/10.1016/j.compag.2024.108894>

Kaczensky, P., Khaliun, S., Payne, J., Boldgiv, B., Buuveibaatar, B., & Walzer, C. (2019). Through the eye of a Gobi khulan—Application of camera collars for ecological research of far-ranging species in remote and highly variable ecosystems. *PLOS ONE*, 14(6), Article e0217772.

<https://doi.org/10.1371/journal.pone.0217772>

Kaler, J., Mitsch, J., Vázquez-Diosdado, J. A., Bollard, N., Dottorini, T., & Ellis, K. A. (2020). Automated detection of lameness in sheep using machine learning approaches: Novel insights into behavioural differences among lame and non-lame sheep. *Royal Society Open Science*, 7(1), 190824.

<https://doi.org/10.1098/rsos.190824>

Kays, R., Crofoot, M. C., Jetz, W., & Wikelski, M. (2015). Terrestrial animal tracking as an eye on life and planet. *Science*, 348(6240), aaa2478. <https://doi.org/10.1126/science.aaa2478>

Kays, R., Davidson, S. C., Berger, M., Bohrer, G., Fiedler, W., Flack, A., Hirt, J., Hahn, C., Gauggel, D., Russell, B., Kölzsch, A., Lohr, A., Partecke, J., Quetting, M., Safi, K., Scharf, A., Schneider, G., Lang, I., Schaeuffelhut, F., ... Wikelski, M. (2022). The Movebank system for studying global animal movement and demography. *Methods in Ecology and Evolution*, 13(2), 419–431.

<https://doi.org/10.1111/2041-210X.13767>

Knight, C. W., Bailey, D. W., & Faulkner, D. (2018). Low-cost global positioning system tracking collars for use on cattle. *Rangeland Ecology & Management*, 71(4), 506–508.

<https://doi.org/10.1016/j.rama.2018.03.002>

Krondorf, M., Bittner, S., Plettemeier, D., Knopp, A., & Wikelski, M. (2022). ICARUS—Very low power satellite-based IoT. *Sensors*, 22(17), Article 6329. <https://doi.org/10.3390/s22176329>

Lahoz-Monfort, J. J., & Magrath, M. J. L. (2021). A comprehensive overview of technologies for species and habitat monitoring and conservation. *BioScience*, 71(10), 1038–1062.

<https://doi.org/10.1093/biosci/biab073>

Lamanna, M., Bovo, M., Bellisola, G., Romanzin, A., & Cavallini, D. (2025). Rethinking wearable technology in dairy cows: Challenges and prospects for smart collars. *The Open Agriculture Journal*, 19. <http://dx.doi.org/10.2174/0118743315410860250914045935>

Lee, M., & Seo, S. (2021). Wearable wireless biosensor technology for monitoring cattle: A review. *Animals*, 11(10), 2779. <https://doi.org/10.3390/ani11102779>

Leolini, L., Moriondo, M., Brilli, L., Galvagno, M., Bindi, M., Argenti, G., Cammarano, D., Bellini, E., Dibari, C., Wohlfahrt, G., Feigenwinter, I., Dal Prà, A., Dalmonech, D., Collalti, A., Cioccolo, E., Cremonese, E., Filippa, G., Staglianò, N., & Costafreda-Aumedes, S. (2023). GRASSVISTOCK: modeling water fluxes in agro-pastoral systems. 2023 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor) (pp. 199–204). IEEE. <https://doi.org/10.1109/MetroAgriFor58484.2023.10424218>

Lund, S., Jacobsen, J., Nielsen, M., Friis, M., Nielsen, N., Mortensen, N., Skibsted, R., Aaser, M., Staahltoft, S., Bruhn, D., Sonne, C., Alstrup, A., Frikke, J., & Pertoldi, C. (2024). Spatial distribution and hierarchical behaviour of cattle using a virtual fence system. *Animals*, 14(14), 2121. <https://doi.org/10.3390/ani14142121>

Macdonald, D. W., & Amlaner Jr, C. J. (1980). A practical guide to radio tracking. In *A handbook on biotelemetry and radio tracking* (pp. 143-159). Pergamon.

Mancuso, D., Castagnolo, G., Parlato, M., Valenti, F., & Porto, S. M. C. (2023). Low-power networks and GIS analyses for monitoring the site use of grazing cattle. *Computers and Electronics in Agriculture*, 210, 107897. <https://doi.org/10.1016/j.compag.2023.107897>

Manly, B. F. J., McDonald, L. L., Thomas, D. L., McDonald, T. L., & Erickson, W. P. (2002). *Resource selection by animals: Statistical design and analysis for field studies* (2nd ed.). Springer-Verlag.

Mao, A., Huang, E., Wang, X., & Liu, K. (2023). Deep learning-based animal activity recognition with wearable sensors: Overview, challenges, and future directions. *Computers and Electronics in Agriculture*, 211, 108043. <https://doi.org/10.1016/j.compag.2023.108043>

McGranahan, D. A., Geaumont, B., & Spiess, J. W. (2018). Assessment of a livestock GPS collar based on an open-source datalogger informs best practices for logging intensity. *Ecology and evolution*, 8(11), 5649-5660. <https://doi.org/10.1002/ece3.4094>

Mendes, C. S., Bartos, I., Márka, Z., Akay, T., Márka, S., & Mann, R. S. (2015). Quantification of gait parameters in freely walking rodents. *BMC Biology*, 13, Article 50. <https://doi.org/10.1186/s12915-015-0154-0>

Moore, C., Beaman, J., Brice, M., & Burke da Silva, K. (2023). A case report assessing the utility of a low-cost tracking GPS device for monitoring terrestrial mammal movements. *Animal Biotelemetry*, 11(1), 35. [10.1186/s40317-023-00347-w](https://doi.org/10.1186/s40317-023-00347-w).

Mulvenna, C. C., Wilson, R. P., Marks, N. J., Maule, A. G., & Scantlebury, D. M. (2018). The ability of magnetic field sensors to monitor feeding in three domestic herbivores. *PeerJ*, 6, e5489. <https://doi.org/10.7717/peerj.5489>

Oquist, K. A., Strock, J. S., & Mulla, D. J. (2007). Influence of alternative and conventional farming practices on subsurface drainage and water quality. *Journal of Environmental Quality*, 36(4), 1194–1204. <https://doi.org/10.2134/jeq2006.0386>

Ossi, F., Urbano, F., & Cagnacci, F. (2019). Biologging and remote sensing of behavior. In J. C. Choe (Ed.), *Encyclopedia of animal behavior* (2nd ed., Vol. 1, pp. 529–537). Academic Press. <https://doi.org/10.1016/B978-0-12-809633-8.90089-X>

Painter, M. S., Silovský, V., Blanco, J., Holton, M., Faltusová, M., Wilson, R., ... & Ježek, M. (2024). Development of a multisensor biologging collar and analytical techniques to describe high-resolution spatial behavior in free-ranging terrestrial mammals. *Ecology and Evolution*, 14(9), e70264. <https://doi.org/10.1002/ece3.70264>

Painter, M. S., Silovský, V., Blanco, J., Holton, M., Faltusová, M., Wilson, R., Börger, L., Psotta, L., Ramos-Almodóvar, F., Estrada, L., Landler, L., Malkemper, E. P., Hart, V., & Ježek, M. (2024). Development of a multisensor biologging collar and analytical techniques to describe high-resolution spatial behavior in free-ranging terrestrial mammals. *Ecology and Evolution*, 14(9), e70264. <https://doi.org/10.1002/ece3.70264>

- Parlato, M. C., Valenti, F., & Porto, S. M. (2024). GIS-based methodology for tracking the grazing cattle site use. *Heliyon*, 10(13). <https://doi.org/10.1016/j.heliyon.2024.e33166>
- Payne, A. R., Hale, C. M., Davidson, S. C., Kendall-Bar, J. M., & Beltran, R. S. (2026). Towards a minimum reporting standard to promote animal welfare and data quality in biologging research. *Animal Biotelemetry*, 14(1), 1. <https://doi.org/10.1186/s40317-025-00438-w>
- Picco, G. P., Molteni, D., Murphy, A. L., Ossi, F., Cagnacci, F., Corrà, M., & Nicoloso, S. (2015). Geo-referenced proximity detection of wildlife with WildScope: Design and characterization. *Proceedings of the 14th International Conference on Information Processing in Sensor Networks* (pp. 238–249). ACM. <https://doi.org/10.1145/2737095.2737103>
- Putfarken, D., Dengler, J., Lehmann, S., & Härdtle, W. (2008). Site use of grazing cattle and sheep in a large-scale pasture landscape: A GPS/GIS assessment. *Applied Animal Behaviour Science*, 111(1–2), 54–67. <https://doi.org/10.1016/j.applanim.2007.05.012>
- Rivas, A., Chamoso, P., González-Briones, A., & Corchado, J. M. (2018). Detection of cattle using drones and convolutional neural networks. *Sensors*, 18(7), 2048. <https://doi.org/10.3390/s18072048>
- Rozier-Delgado, P., Garel, M., Rousset, C., Loison, A., & Morel, B. (2025). Modelling endurance in free-ranging animals using tracking collars: insights from domestic hunting dogs. *Journal of Experimental Biology*, 228(10), jeb250259. <https://doi.org/10.1242/jeb.250259>
- Rutz, C., Morrissey, M. B., Burns, Z. T., Burt, J., Otis, B., St Clair, J. J. H., & James, R. (2015). Calibrating animal-borne proximity loggers. *Methods in Ecology and Evolution*, 6(6), 656–667. <https://doi.org/10.1111/2041-210X.12370>
- Ryan, K. P., Ferguson, S. H., Koski, W. R., Young, B. G., Roth, J. D., & Watt, C. A. (2022). Use of drones for the creation and development of a photographic identification catalogue for an endangered whale population. *Arctic Science*, 8(4), 1191–1201. <https://doi.org/10.1139/as-2021-0038>
- Schoenecker, K. A., King, S. R. B., Hennig, J. D., Cole, M. J., Scasta, J. D., & Beck, J. L. (2024). Effects of telemetry collars on two free-roaming feral equid species. *PLOS ONE*, 19(5), e0303312. <https://doi.org/10.1371/journal.pone.0303312>
- Shao, W., Kawakami, R., Yoshihashi, R., You, S., Kawase, H., & Naemura, T. (2020). Cattle detection and counting in UAV images based on convolutional neural networks. *International Journal of Remote Sensing*, 41(1), 31–52. <https://doi.org/10.1080/01431161.2019.1624858>
- Simanungkalit, K., Barwick, J., Cowley, F. C., Dobos, R. C., & Hegarty, R. S. (2021). Validation of an accelerometer-based sensor for classifying grazing behavior of sheep. *Applied Animal Behaviour Science*, 244, 105491. <https://doi.org/10.1016/j.applanim.2021.105491>
- Stewart, D. G., Mendes, E. D. M., Lee, K., Blum, M. E., Tedeschi, L. O., & Webb, S. L. (2025). Comparison of GPS collars and solar-powered GPS ear tags for animal movement studies. *Smart Agricultural Technology*, 11, Article 101021. <https://doi.org/10.1016/j.atech.2025.101021>
- Swain, D. L., Friend, M. A., Bishop-Hurley, G. J., Handcock, R. N., & Wark, T. (2011). Tracking livestock using global positioning systems—are we still lost? *Animal Production Science*, 51(3), 167–175. <https://doi.org/10.1071/AN10149>
- Tuytens, F. A., Molento, C. F., & Benaissa, S. (2022). Twelve threats of precision livestock farming (PLF) for animal welfare. *Frontiers in Veterinary Science*, 9, 889623. <https://doi.org/10.3389/fvets.2022.889623>

- Ungar, E. D., Henkin, Z., Gutman, M., Dolev, A., Genizi, A., & Ganskopp, D. (2005). Inference of animal activity from GPS collar data on free-ranging cattle. *Rangeland Ecology & Management*, 58(3), 256–266. [https://doi.org/10.2111/1551-5028\(2005\)58\[256:IOAAFG\]2.0.CO;2](https://doi.org/10.2111/1551-5028(2005)58[256:IOAAFG]2.0.CO;2)
- Urbano, F., & Cagnacci, F. (2021). Data management and sharing for collaborative science: Lessons learnt from the Euromammals initiative. *Frontiers in Ecology and Evolution*, 9, Article 727023. <https://doi.org/10.3389/fevo.2021.727023>
- Versluijs, E., Niccolai, L., Spedener, M., Zimmermann, B., Hesse, A., Tofastrud, M., Devineau, O., & Evans, A. L. (2023). Classification of behaviors of free-ranging cattle using accelerometry signatures collected by virtual fence collars. *Frontiers in Animal Science*, 4, 1083272. <https://doi.org/10.3389/fanim.2023.1083272>
- Vidal-Cardos, R., Fàbrega, E., & Dalmau, A. (2025). Activity and behavior patterns of cattle, horses, and sheep grazing in mountainous areas using geolocation collars. *Journal of Animal Science*, 103, skaf318. <https://doi.org/10.1093/jas/skaf318>
- Walton, E., Casey, C., Mitsch, J., Vázquez-Diosdado, J. A., Yan, J., Dottorini, T., Ellis, K. A., Winterlich, A., & Kaler, J. (2018). Evaluation of sampling frequency, window size and sensor position for classification of sheep behaviour. *Royal Society Open Science*, 5(2), 171442. <https://doi.org/10.1098/rsos.171442>
- Wang, L., Arablouei, R., Alvarenga, F. A., & Bishop-Hurley, G. J. (2023). Classifying animal behavior from accelerometry data via recurrent neural networks. *Computers and Electronics in Agriculture*, 206, 107647. <https://doi.org/10.1016/j.compag.2023.107647>
- Waterhouse, T. (2023). Virtual fencing systems: balancing production and welfare outcomes. *Livestock*, 28(5), 227–234. <https://doi.org/10.12968/live.2023.28.5.227>
- Wild, T. A., van Schalkwyk, L., Viljoen, P., Heine, G., Richter, N., Vorneweg, B., Koblit, J. C., Dechmann, D. K. N., Rogers, W., Partecke, J., Linek, N., Volkmer, T., Gregersen, T., Havmøller, R. W., Morelle, K., Daim, A., Wiesner, M., Wolter, K., Fiedler, W., ... Wikelski, M. (2023). A multi-species evaluation of digital wildlife monitoring using the Sigfox IoT network. *Animal Biotelemetry*, 11, Article 13. <https://doi.org/10.1186/s40317-023-00326-1>
- Wilmers, C. C., Nickel, B., Bryce, C. M., Smith, J. A., Wheat, R. E., & Yovovich, V. (2015). The golden age of bio-logging: How animal-borne sensors are advancing the frontiers of ecology. *Ecology*, 96(7), 1741–1753. <https://doi.org/10.1890/14-1401.1>
- Zhang, H., Sun, Y., Zhao, C., Wang, B., Li, B., & Wang, B. (2023). Review on typical behavior monitoring and physiological condition identification methods for ruminant livestock. *Transactions of the Chinese Society for Agricultural Machinery*, 54(3), 1–21.

List of abbreviations and definitions

Abbreviations	Definitions
ACC	Accelerometer
ADD	Additional, voluntary tagging
AMS	Area Monitoring System
B	Biologger
CAP	Common Agricultural Policy
FSK	Frequency-Shift Keying
GDPR	General Data Protection Regulation
GNSS	Global Navigation Satellite Systems
GPS	Global Position System
GSM	Global System for Mobile Communications
GYR	Gyroscope
IACS	Integrated Administration and Control System
ICARUS	International Cooperation for Animal Research Using Space
IMU	Inertial Measurement Unit
IoT	Internet of Things
IRIDIUM	Iridium Communications Inc. (Satellite telecommunications)
LPWAN	Low-Power Wide-Area Network
MAG	Magnetometer
NOT	Not permitted
ORD	Ordinary, compulsory tagging
PEFL	Precision Extensive Livestock Farming

Abbreviations	Definitions
PLF	Precision Livestock Farming
PIT tag	Passive Integrated Transponders tag
RFID	Radio Frequency Identification
Rs	Remote sensor
RSSI	Received Signal Strength Indicator
TRACES	TRAdE Control and Expert System
UAV	Unmanned Aerial Vehicle
UHF	Ultra High Frequency
VHF	Very High Frequency
WSN	Wireless Sensor Network

List of figures

Figure 1. Biologging systems for animal detection and live tracking.....	18
Figure 2. Recent uptake of literature on PLEF as reported by a recent review performed with systematic criteria within the period 1997–September 2024.....	21
Figure 3. Example of biologging network deployed on livestock and farming systems (either PLF or PEFL).	27
Figure 4. Example of communication pathways from PEFL biologgers to end users.	27
Figure 5. Examples of smart ear and leg tags deployed on livestock. From top left: cows wearing a collar (a) and an ear tag (b) used for data collection on behaviour. Sheep wearing Inertial Measurement Unit sensors (c). CERES TAG, a commercial solar-powered satellite tracking device (d) developed for rangeland animals, and an application on the field (e). MOOvement, a commercial solar-powered tag relying on LPWAN developed for cows (f), and an application in the field (g).....	28

List of tables

Table 1. Authorized means of identification for kept animals under Commission Delegated Regulation (EU) 2019/2035, with specific derogations available.....	7
Table 2. Potential added value of livestock movement tracking with biologging devices mapped against the objectives defined in the Strategic Dialogue on the future of EU agriculture.....	11
Table 3. Regulatory status of external electronic identifiers (ear tags and pastern bands) by domestic animal species.....	13
Table 4. Requirement analysis for the deployment of biologgers. Literature references in Notes...	15
Table 5. Main biologging sensors and technology for basic and applied studies of wild terrestrial mammals.	19
Table 6. Recent reviews on biologging use in the PLF domain.....	22
Table 7. Main biologging sensors and technology for livestock identification and tracking (mainly PLEF).	24
Table 8. Available PLEF market products as per a systematic review.	29
Table 9. Potential use of biologging sensors associated with external electronic identifiers (ear tags and pastern bands) for livestock (extensive/hybrid: PLEF).....	31
Table 10. Potential for leveraging existing projects and initiatives for introducing biologging for farmed animals.....	35

Getting in touch with the EU

In person

All over the European Union there are hundreds of Europe Direct centres. You can find the address of the centre nearest you online (european-union.europa.eu/contact-eu/meet-us_en).

On the phone or in writing

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

- by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),
- at the following standard number: +32 22999696,
- via the following form: european-union.europa.eu/contact-eu/write-us_en.

Finding information about the EU

Online

Information about the European Union in all the official languages of the EU is available on the Europa website (europa.eu).

EU publications

You can view or order EU publications at op.europa.eu/en/publications. Multiple copies of free publications can be obtained by contacting Europe Direct or your local documentation centre (european-union.europa.eu/contact-eu/write-us_en).

EU law and related documents

For access to legal information from the EU, including all EU law since 1951 in all the official language versions, go to EUR-Lex (eur-lex.europa.eu).

Open data from the EU

The portal data.europa.eu provides access to open datasets from the EU institutions, bodies and agencies. These can be downloaded and reused for free, for both commercial and non-commercial purposes. The portal also provides access to a wealth of datasets from European countries.

Science for policy

The Joint Research Centre (JRC) provides independent, evidence-based knowledge and science, supporting EU policies to positively impact society



Scan the QR code to visit:

[Joint Research Centre](https://joint-research-centre.ec.europa.eu)

<https://joint-research-centre.ec.europa.eu>



Publications Office
of the European Union